Ocular Embryology

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At what age does the eye start to develop ?

- 3 wks
- 6 wks
- 2 months
- 3 months

- Important elements in genesis of the eye:
  1. growth factors
  2. homeobox genes
  3. neural crest cells

Growth factors

- Induce ocular development by communication through chemical signals
- Active in the earliest stages of embryo
- Include:
  - Fibroblast growth factor (FGF)
  - Transforming growth factor β (TGF- β1 and TGF- β2)
  - Insulin-like growth factor I (IGF-I)

Growth factors

- Direct migration and developmental patterns of cranial neural crest cells during morphogenesis of optic vesicle and lens

- Control differentiation of various ocular tissues

- Regulate level of expression of homeobox genes

Homeobox genes

- Distinctive segment of DNA (180 base pairs)
- Master genes: control many subordinate genes
- Conserved evolutionarily: present through plant and animal kingdoms
- Activated by GF and also retinoic acid

(too much or too little vit A is teratogenic)

Homeobox genes

- Specific homeobox genes are involved in development of the eye
- Ex. PAX6 gene, HOX8.1, HOX7.1
- homeobox genes control the overall arrangement of the eye as an organ
- visual acuity requires precise spatial arrangement of tissues of the eye
- homeobox genes must be expressed at the appropriate level and time

Neural crest cells

- arise from neuroectoderm located at the crest of the neural folds at approximately the same time that the folds fuse to form the neural tube

e CNS (brain+spinal cord

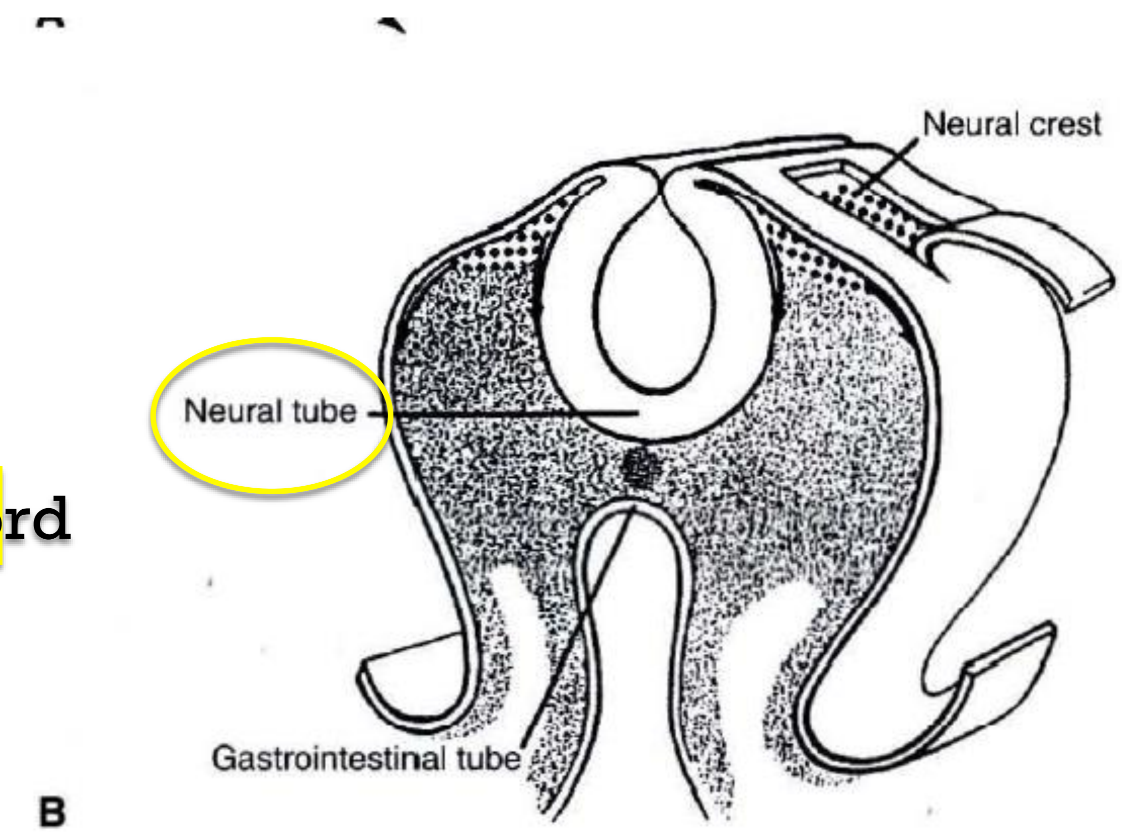


Figure 4-3 Cross sections through embryos before **(A)** and after **(B)** the onset of migration of crest cells *(diamond pattern)*. The ectoderm has been peeled back in **B** to show the underlying neural crest cells. *(Reproduced with permission from Johnston MC, Sulik KK. Development of face and oral cavity. In: Bhaskar SN, ed.* Orban's Oral Histology and Embryology. *11th ed. St Louis: Mosby; 1991.)*

Neural crest cells

- they migrate to different regions of the embryo and differentiation occurs
- Most mesenchymal cells of the facial primordia are derived from the neural crest.

Neural crest cells

- Early in development, neural crest cells are pleuripotent, and their final differentiation is influenced by local factors

Neural crest cells

- Neural crest cells make a major contribution to the connective tissue components of the eye and orbit.
- Notable exceptions :
  - striated fibers of the extraocular muscles
  - endothelial cells of all blood vessels of the eye and orbit.

  (arise from mesoderm)

Table 4-1 Derivatives of Embryonic Tissues

ECTODERM

Neuroectoderm

- Neurosensory retina
- Retinal pigment epithelium
- Pigmented ciliary epithelium
- Nonpigmented ciliary epithelium
- Pigmented iris epithelium
- Sphincter and dilator muscles of iris
- Optic nerve, axons, and glia
- Vitreous

Cranial Neural Crest Cells

- Corneal stroma and endothelium
- Sclera (see also mesoderm)
- Trabecular meshwork
- Sheaths and tendons of extraocular muscles
- Connective tissues of iris
- Ciliary muscles
- Choroidal stroma
- Melanocytes (uveal and epithelial)
- Meningeal sheaths of the optic nerve
- Schwann cells of ciliary nerves
- Ciliary ganglion
- All midline and inferior orbital bones, as well as parts of orbital roof and lateral rim
- Cartilage
- Connective tissue of orbit
- Muscular layer and connective tissue sheaths of all ocular and orbital vessels

Surface Ectoderm

- Epithelium, glands, cilia of skin of eyelids and caruncle
- Conjunctival epithelium
- Lens
- Lacrimal gland
- Lacrimal drainage system
- Vitreous

MESODERM

- Fibers of extraocular muscles
- Endothelial lining of all orbital and ocular blood vessels
- Temporal portion of sclera
- Vitreous

BCSC section 2 ch.4

Neurocristopathy

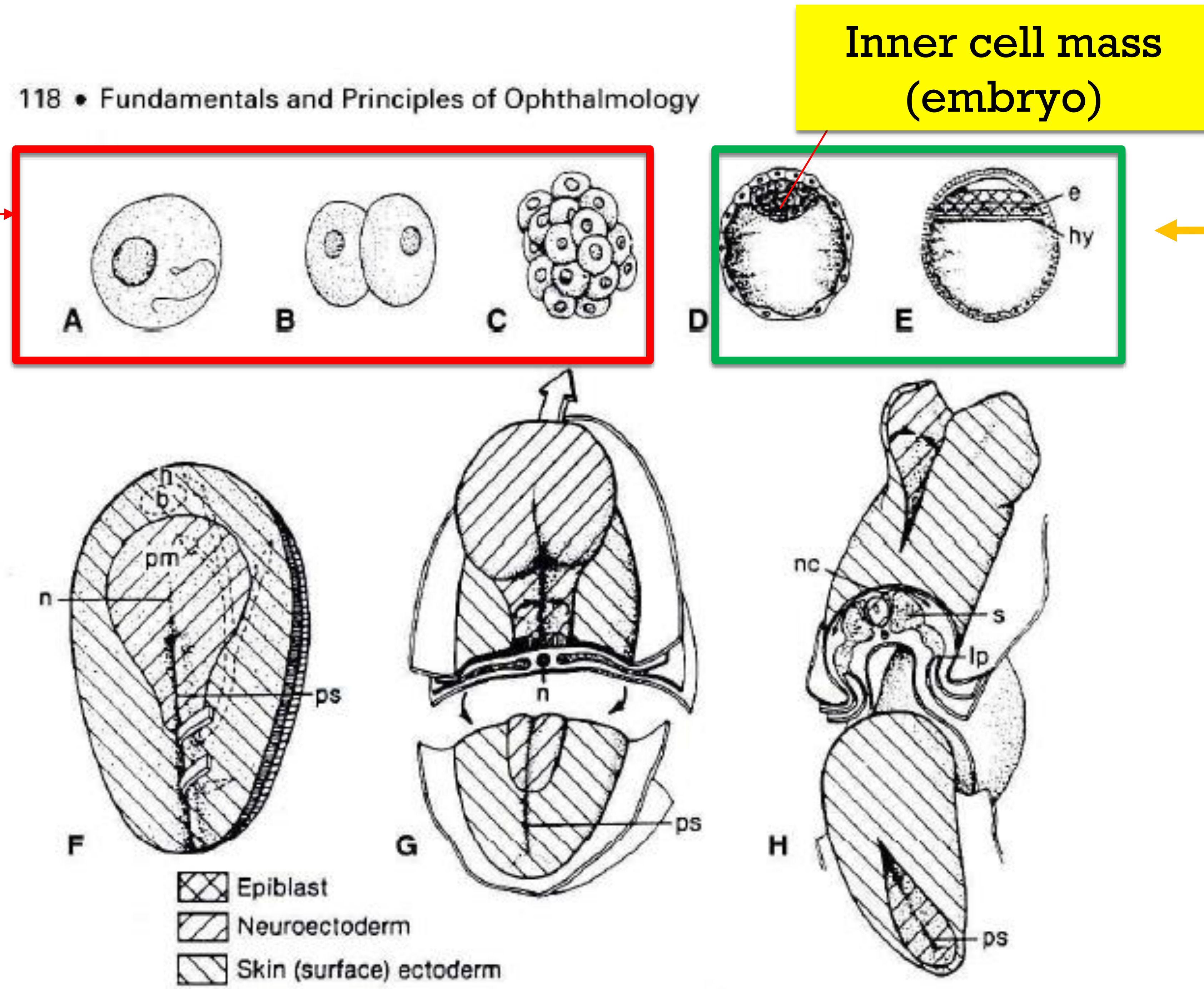
- Congenital and developmental anomalies that involve cells derived from the neural crest have been grouped together under the term neurocristopathies.
- From defects in migration or terminal differentiation
- Ex: craniofacial malformations

- Embryogenesis
- Organogenesis

Embryogenesis

Embryogenesis

Morula



Blastula
-epiblast
-hypoblast

- First 2 wks after fertilization:
- Morula □ blastula □ gastrula phase

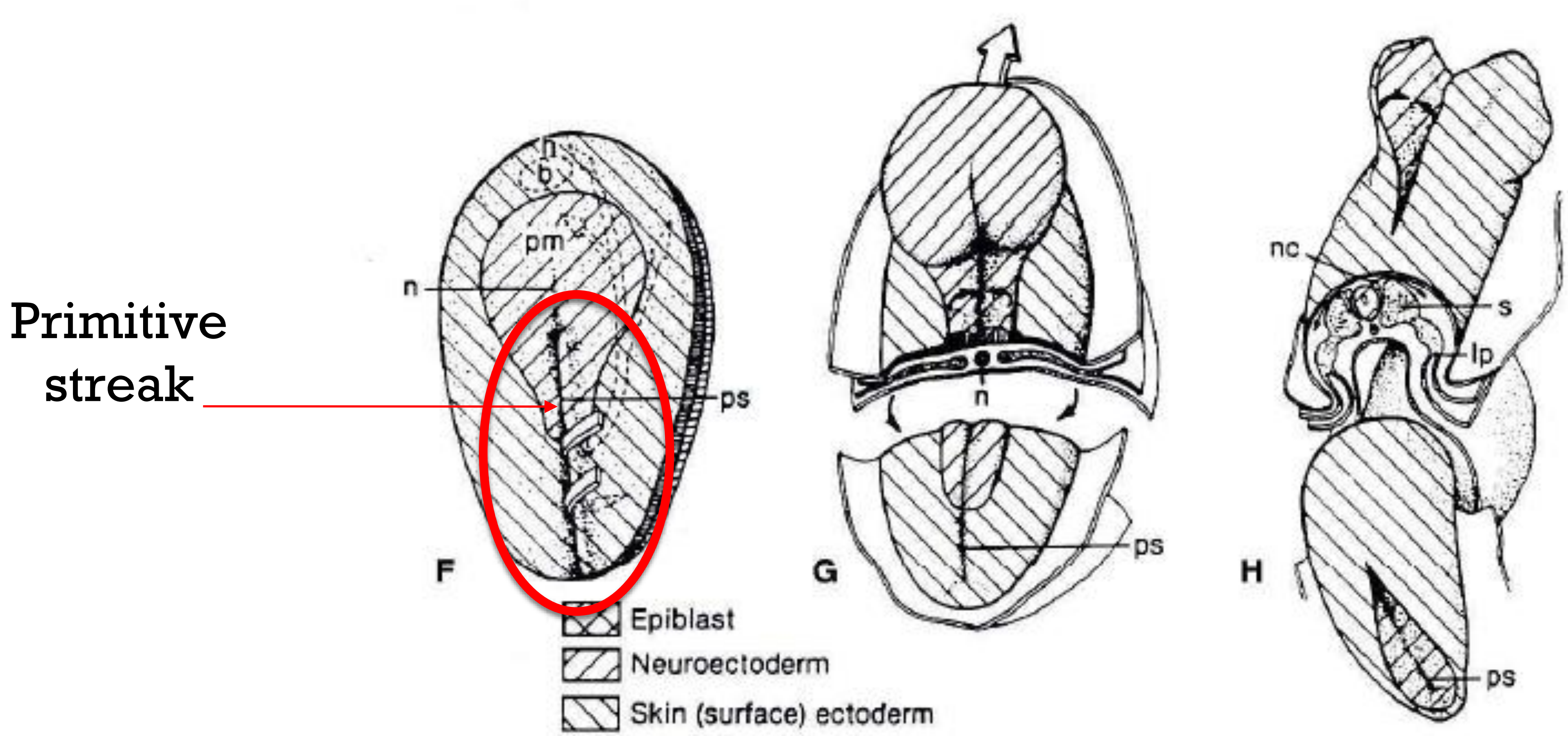
2 wks after fertilisation: morula -> blastula -> gastrula

Gastrulation

- process that results in the establishment of the 3 primary germ layers:
  - Ectoderm
  - Mesoderm
  - Endoderm

Gastrulation

- Cells of the epiblast in the medial region of the embryonic disc begin to proliferate at the caudal (หาง) end which causes the development of a thickening known as the primitive streak.

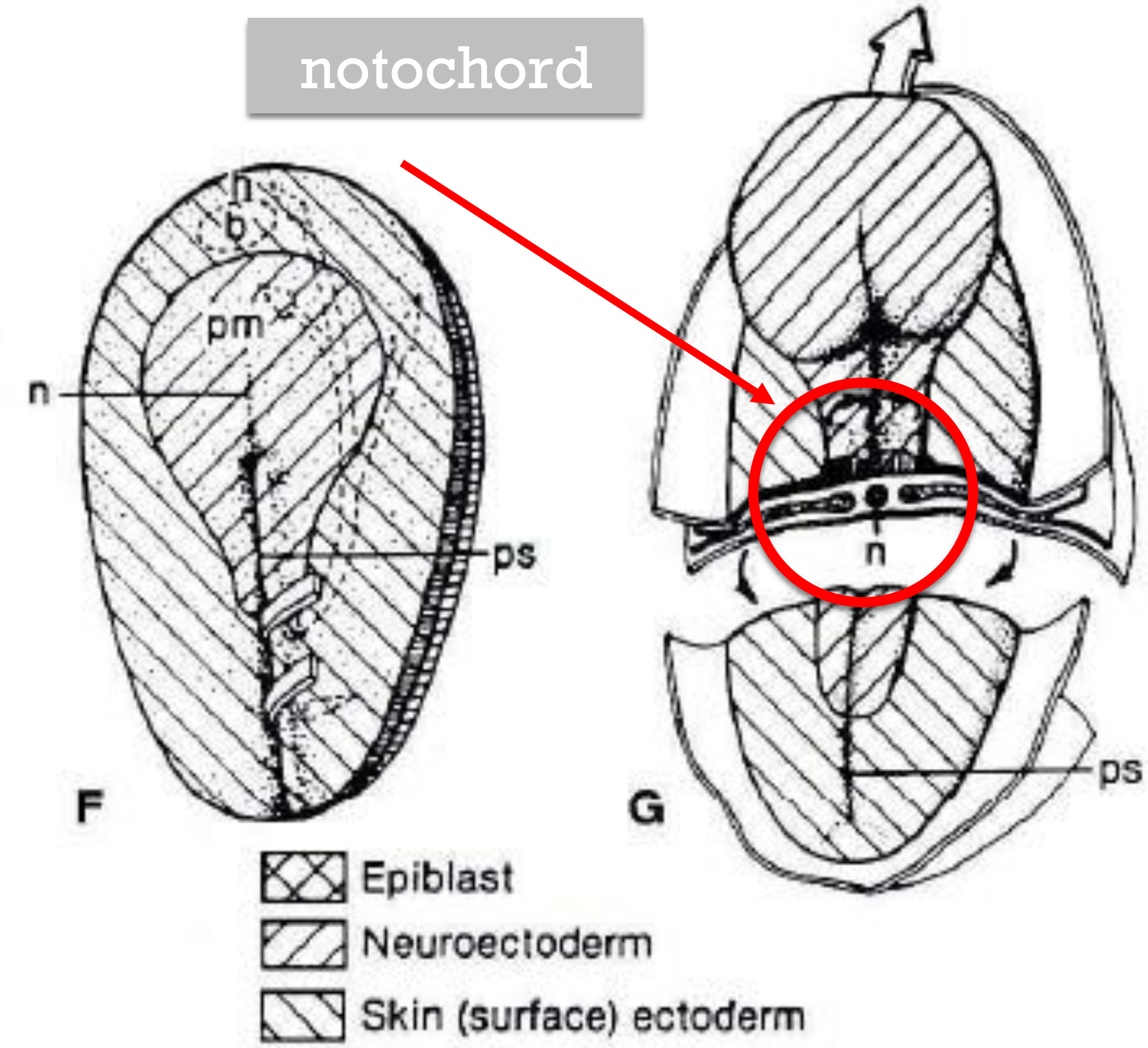


- cells of the primitive streak migrate both laterally and cephalad beneath the epiblast -> mesoderm
- cells that remain in the epiblast are now recognized as the embryonic ectoderm
- Some cells of the primitive streak invade the hypoblast and laterally displace most of these cells to give rise to the embryonic endoderm

- primitive streak elongates at the caudal end

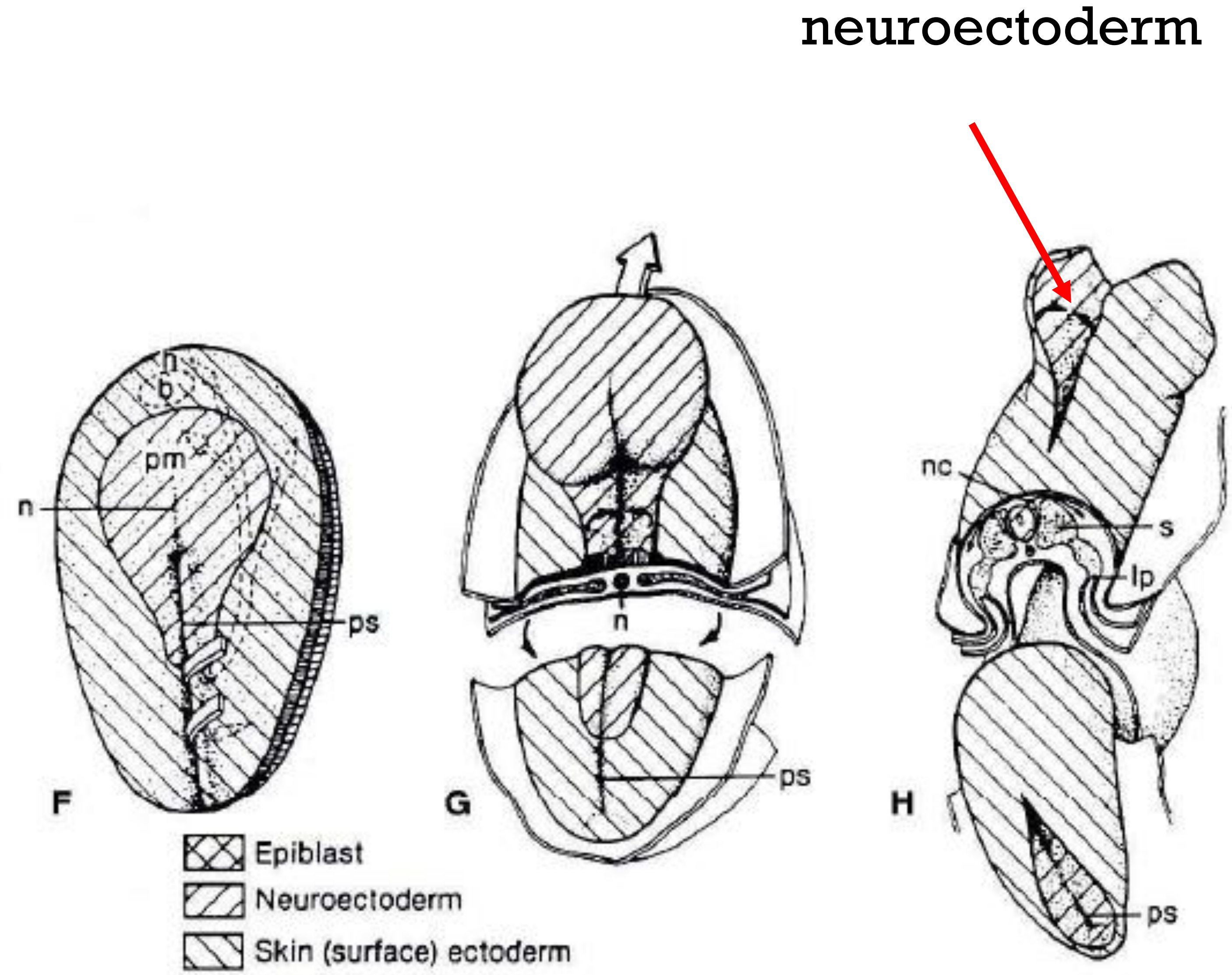
- The cranial end of the primitive streak enlarges as the primitive node.

  which develops into axial skeleton (or notochord) of the embryo.



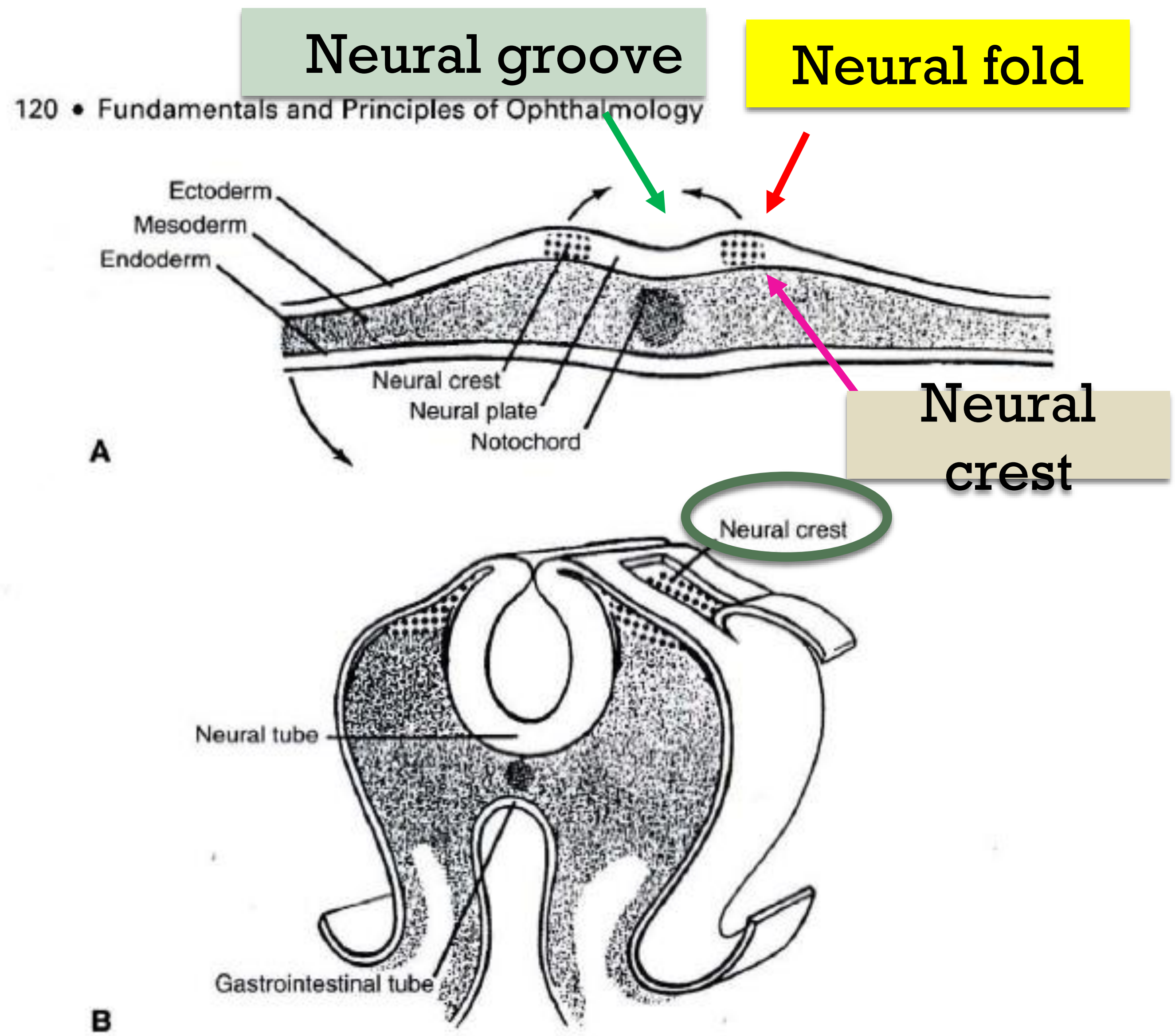
- The development of the notochord induces the overlying ectoderm to differentiate into neuroectoderm (neural plate)

- The brain and the eye develop from the most anterior region of the neural plate.



- Growth of the lateral part of the neural plate results in folds that develop upward and outward, parallel to the neural groove from the head to the caudal region

- The neuroectodermal cells at the apex of the folds proliferate and produce a population of neural crest cells, which contribute extensively to the tissues of the eye



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Organogenesis of the eye

Organogenesis of the eye

- Optic sulcus = indentation in each neural fold
- Evagination of sulcus -> Optic pit (single layer of neuroectoderm)
- As the neural tube closes, the pits deepen and become optic vesicles

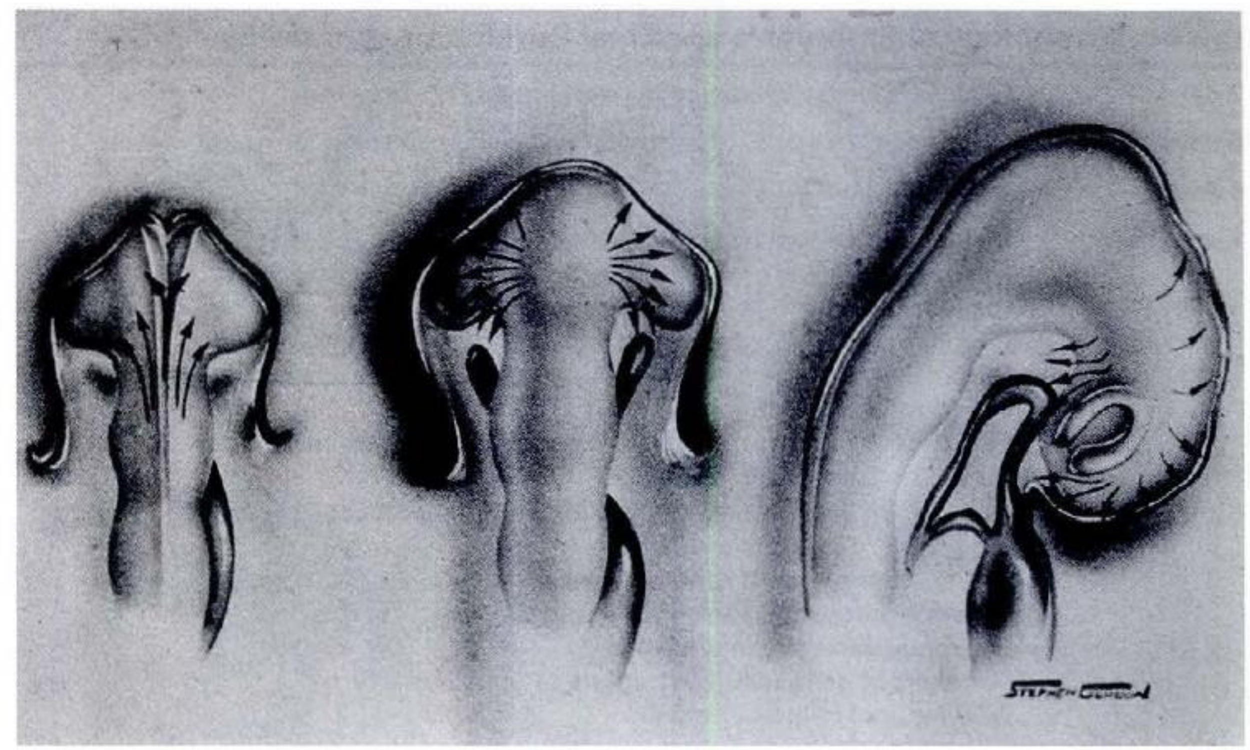
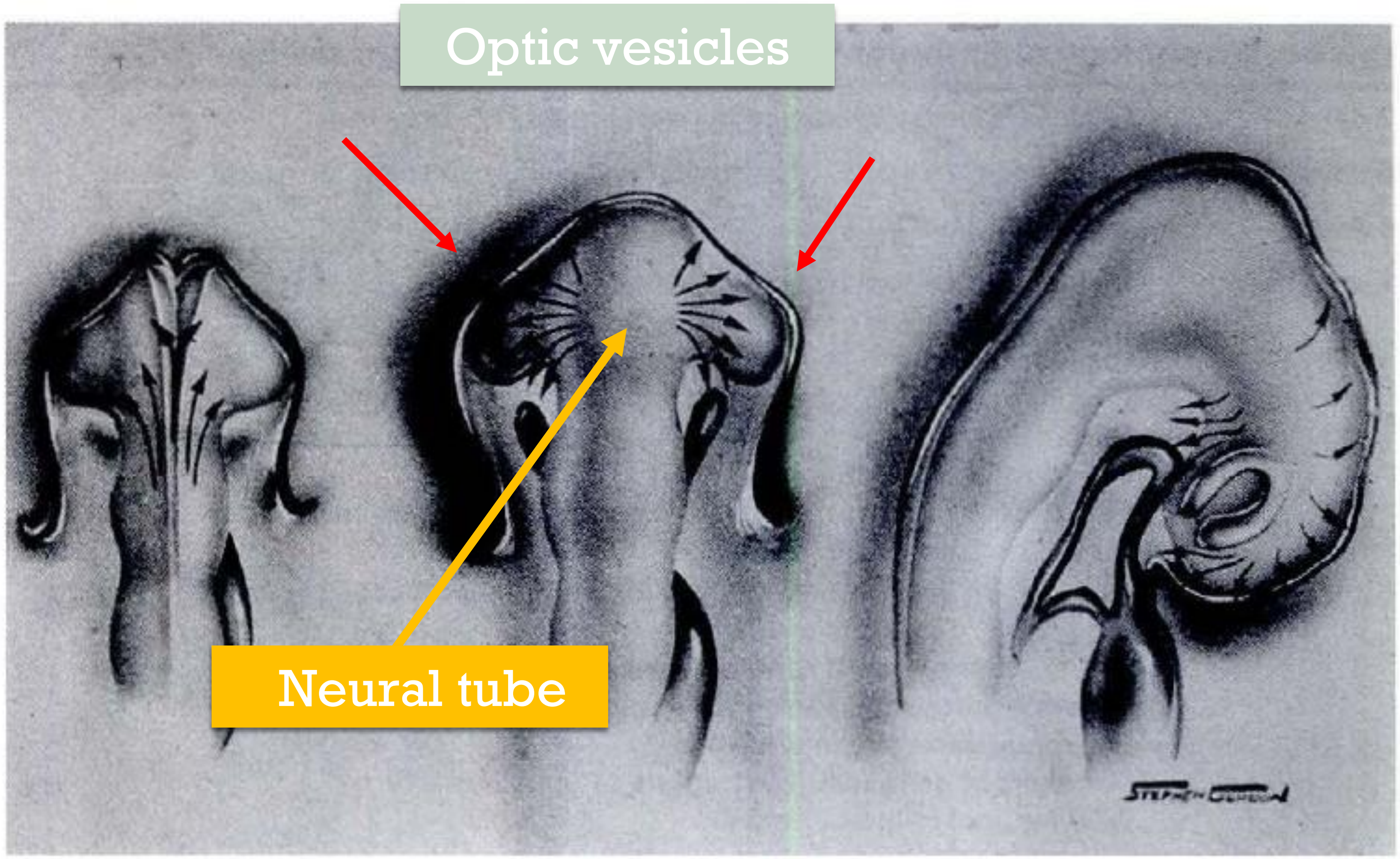


Figure 4-4 Migration of cranial neural crest cells from dorsal diencephalic and mesencephalic regions. **Left,** Cells begin migration anteriorly as tube closes. **Center,** Crest cells move in waves around the optic vesicle and lose continuity with the surface cells. **Right,** The neural tube flexes ventrally, carrying the optic cup and crest cells ventrally. *(Redrawn from M. Johnston, 1966.)*



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Lined with neuroectoderm
al cells

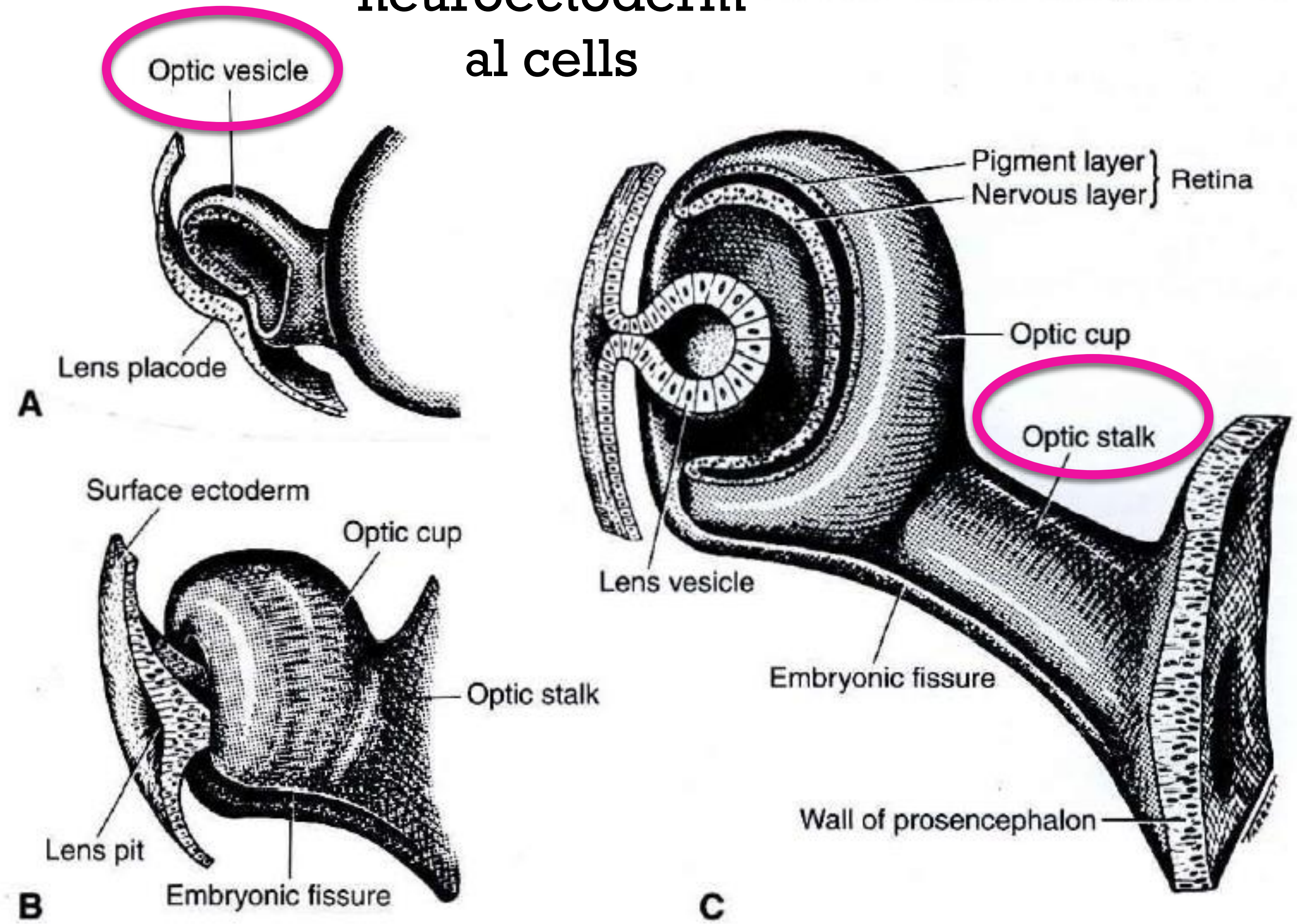
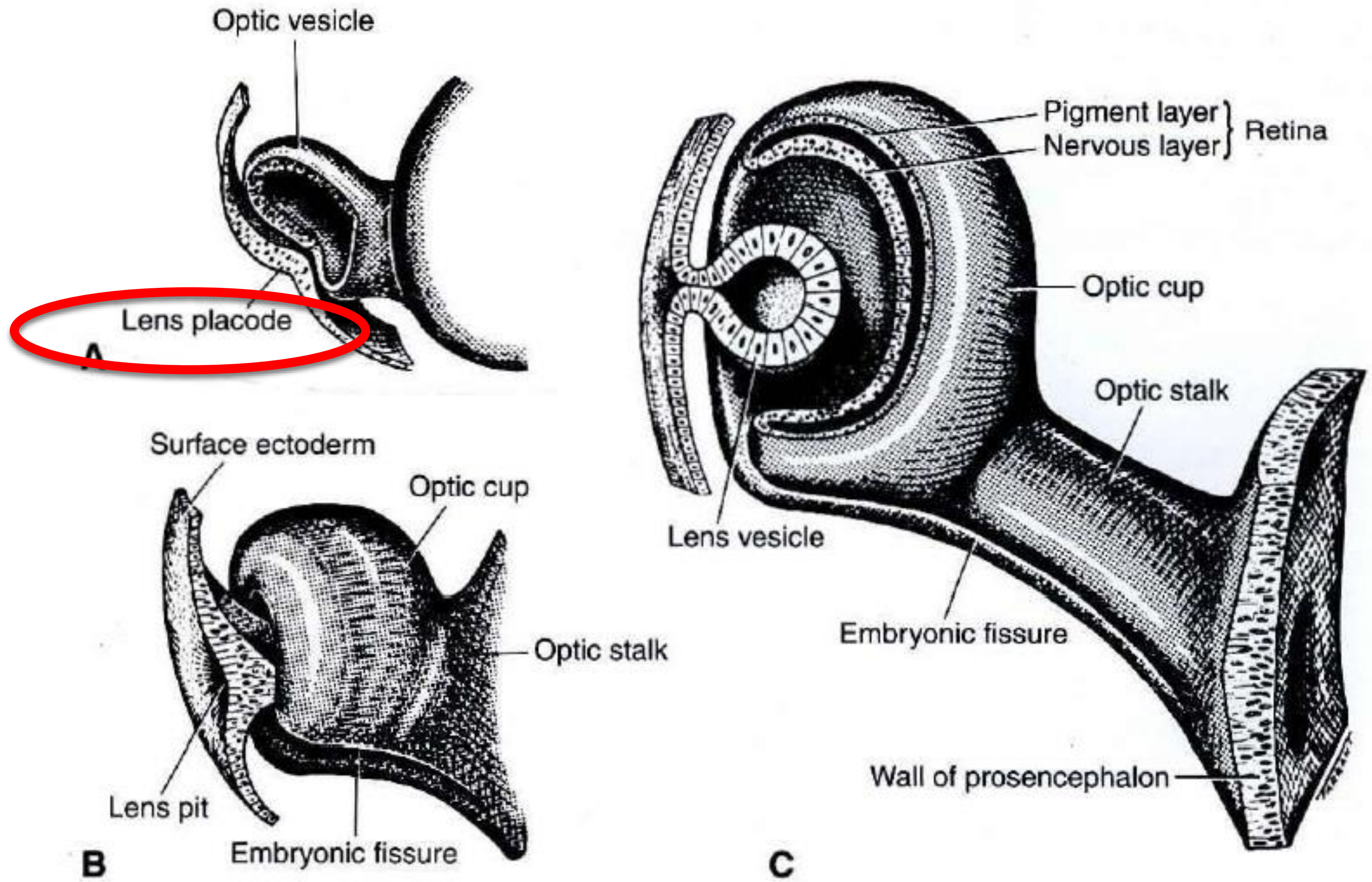


Figure 4-5 Diagram of the development of the human optic cup. The optic vesicle and cup

The optic vesicles remain attached to, and continuous with, the neural tube (ส่วนที่เป็น forebrain vesicle) by optic stalks composed of neuroectodermal cells



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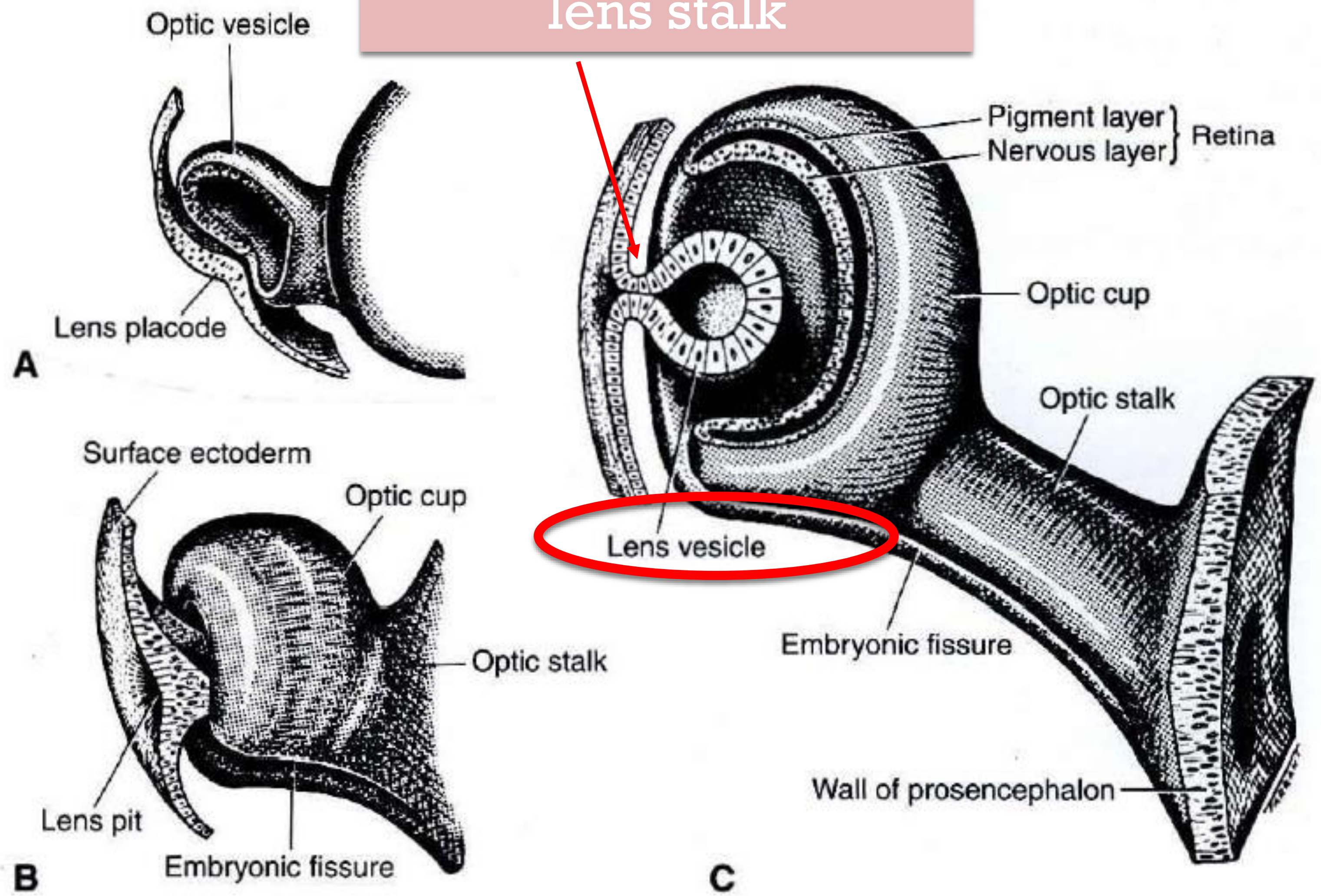
As the optic vesicle approaches the outer wall (ผิวๆ) of the embryo, a focal thickening of the cells, the lens placode, develops in the surface ectoderm

Lens vesicle attached to surface ectoderm by lens stalk



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4th wk: invagination of the lens placode --> formation of the lens vesicle, which initially remains attached to the surface ectoderm by the lens stalk

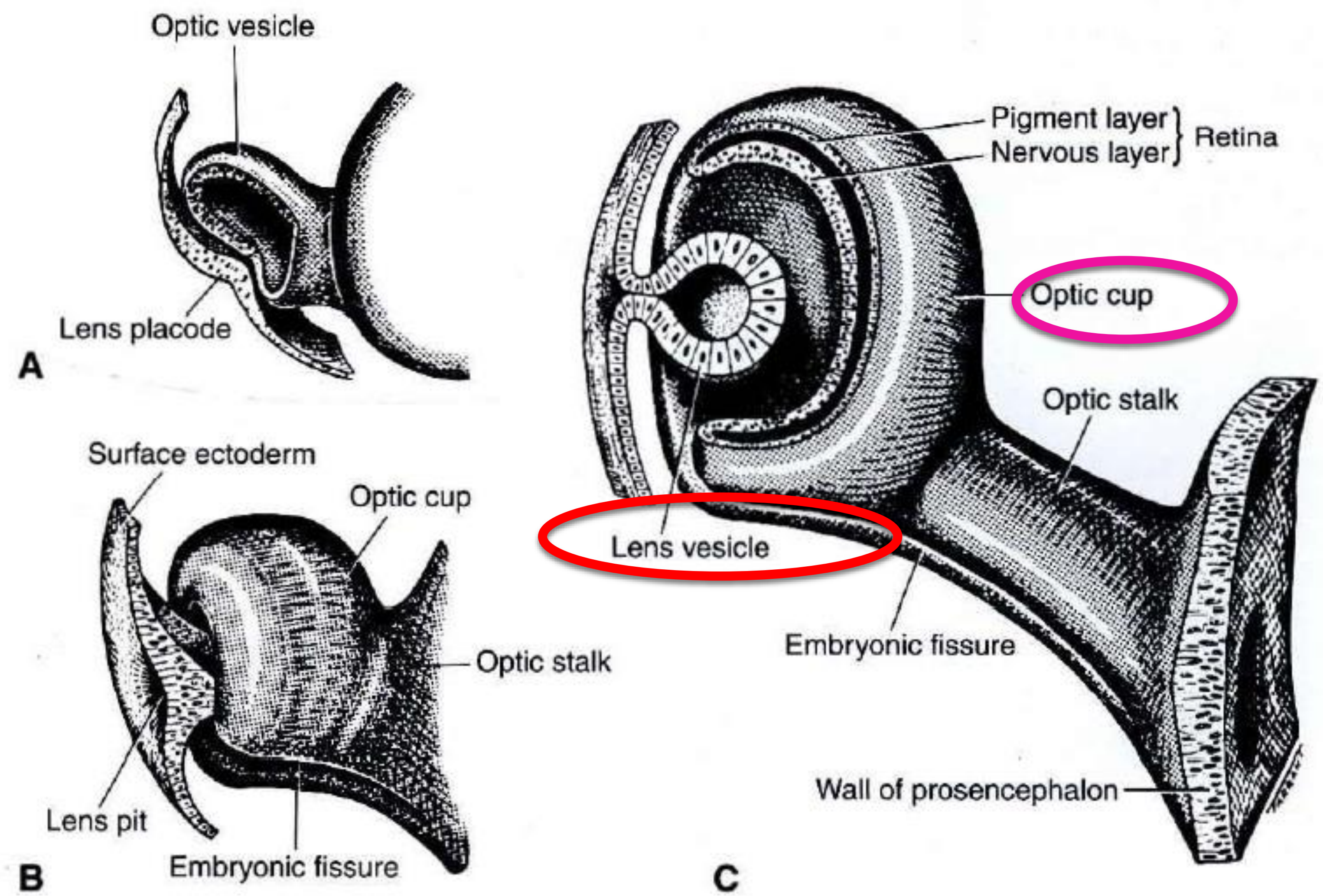


Figure 4-5 Diagram of the development of the human optic cup. The optic vesicle and cup

Simultaneously, invagination optic vesicle (temporal and lower walls) -> formation of the optic cup.

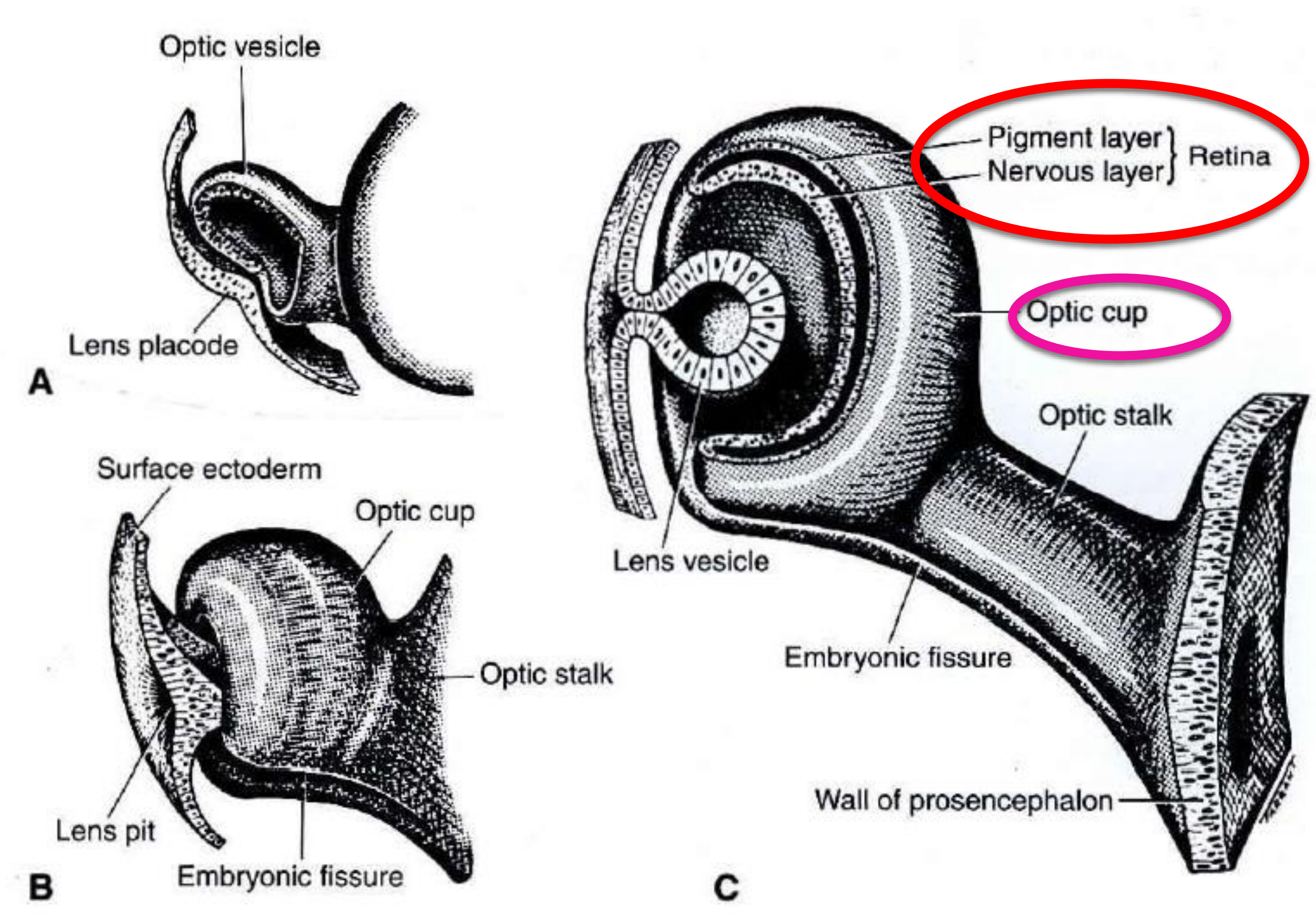


Figure 4-5 Diagram of the development of the human optic cup. The optic vesicle and cup

The outer layer of the optic cup (monolayer of cells) -> retinal pigment epithelium (RPE). The inner, invaginated layer neurosensory retina

Initially, the cup is incomplete in its inferior portion

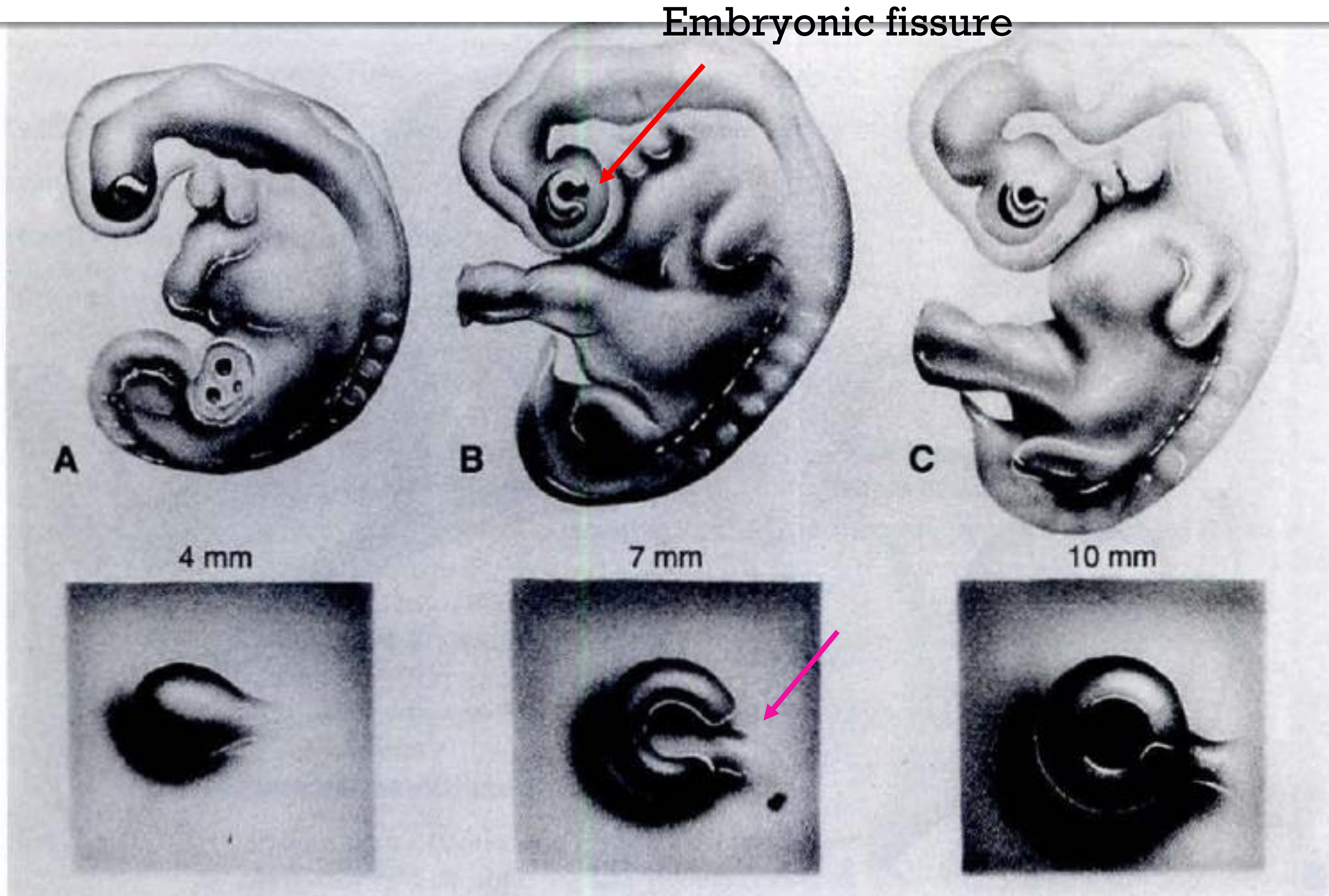


Figure 4-6 Ocular and somatic development. **A,** Flexion of the neural tube and ballooning of the optic vesicle. **B,** Upper-limb buds appear as the optic cup and embryonic fissure emerge. **C,** Completion of the optic cup with closure of the fissure. Convolutions appear in the brain, and leg buds appear. The size of the fetus is given. *Lower sequence:* optic vesicle; optic cup with open embryonic fissure; cup with fissure closing.

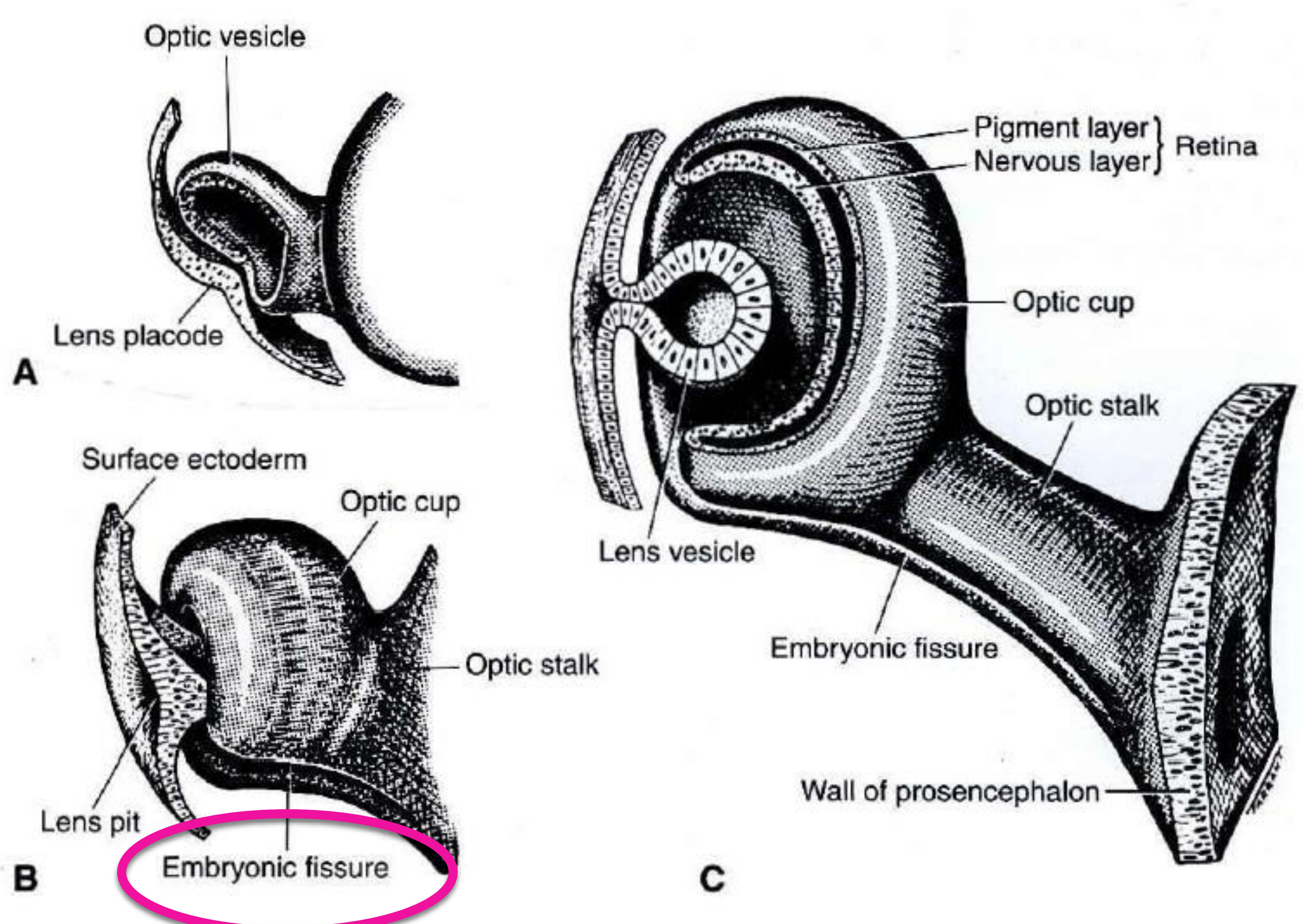


Figure 4-5 Diagram of the development of the human optic cup. The optic vesicle and cup

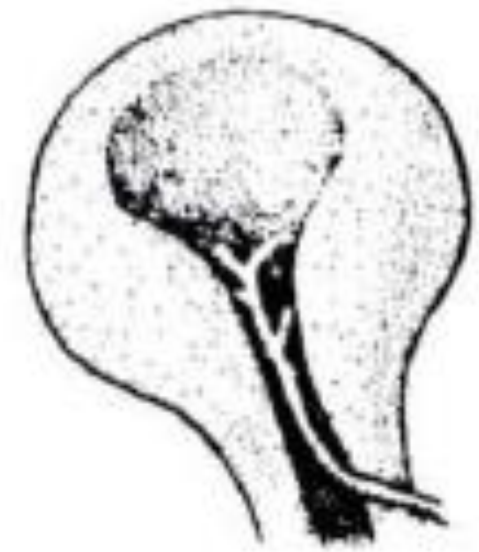
The fissure between the folds or margins of the cup is called the embryonic fissure (previously the choroidal, or fetal, fi ssure) เกิดจาก invagination ของ neuroectoderm

Table 4-2 Chronology of Embryonic and Fetal Development of the Eye

22 days	Optic primordium appears in neural folds (1.5–3.0 mm).
25 days	Optic vesicle evaginates. Neural crest cells migrate to surround vesicle.
28 days	Vesicle induces lens placode.
Second month	Invagination of optic and lens vesicles. Hyaloid artery fills embryonic fissure. Closure of embryonic fissure begins. Pigment granules appear in retinal pigment epithelium. Primordia of lateral rectus and superior oblique muscles grow anteriorly. Eyelid folds appear. Retinal differentiation begins with nuclear and marginal zones. Migration of retinal cells begins. Neural crest cells of corneal endothelium migrate centrally. Corneal stroma follows. Cavity of lens vesicle is obliterated. Secondary vitreous surrounds hyaloid system. Choroidal vasculature develops. Axons from ganglion cells migrate to optic nerve. Glial laminal cribrosa forms. Bruch's membrane appears.
Third month	Precursors of rods and cones differentiate. Anterior rim of optic vesicle grows forward, and ciliary body starts to develop. Sclera condenses. Vortex veins pierce sclera. Eyelid folds meet and fuse.
Fourth month	Retinal vessels grow into nerve fiber layer near optic disc. Folds of ciliary processes appear. Iris sphincter develops. Descemet's membrane forms. Schlemm's canal appears. Hyaloid system starts to regress. Glands and cilia develop.
Fifth month	Photoreceptors develop inner segments. Choroidal vessels form layers. Iris stroma is vascularized. Eyelids begin to separate.
Sixth month	Ganglion cells thicken in macula. Recurrent arterial branches join the choroidal vessels. Dilator muscle of iris forms.
Seventh month	Outer segments of photoreceptors differentiate. Central fovea starts to thin. Fibrous lamina cribrosa forms. Choroidal melanocytes produce pigment. Circular muscle forms in ciliary body.
Eighth month	Chamber angle completes formation. Hyaloid system disappears.
Ninth month	Retinal vessels reach the periphery. Myelination of fibers of optic nerve is complete to lamina cribrosa. Pupillary membrane disappears.

BCSC section 2: ch 4.

Figure 4-7 Optic cup and stalk with open embryonic fissure below. The hyaloid artery from the dorsal ophthalmic artery enters the cavity through the posterior aspect of the embryonic fissure. The rim of the optic cup is above. The lens is not shown.



BCSC section

- The embryonic fissure extends from the rim of the cup near the lens to the distal optic stalk.
- This fissure allows vessels of the hyaloid system to be incorporated within the eye

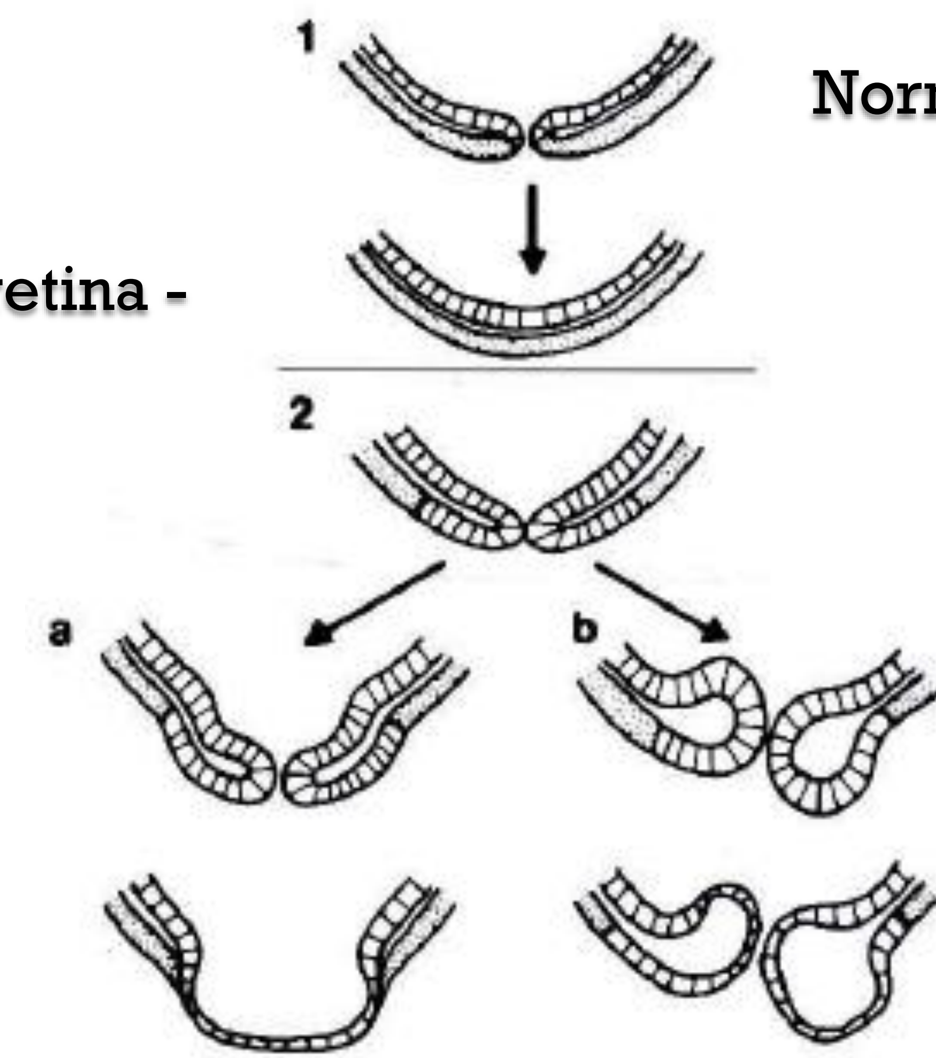
Embryonic fissure

- Closure begins in the midregion of the cup near the equator of the globe and proceeds anteriorly to the rim
- and posteriorly down the stalk, enclosing the hyaloid artery
- Incomplete or inadequate closure produces a coloboma of the iris, ciliary body, choroid, or optic disc, depending on the extent of the failed closure

Closure of embryonic fissure

Normal closure

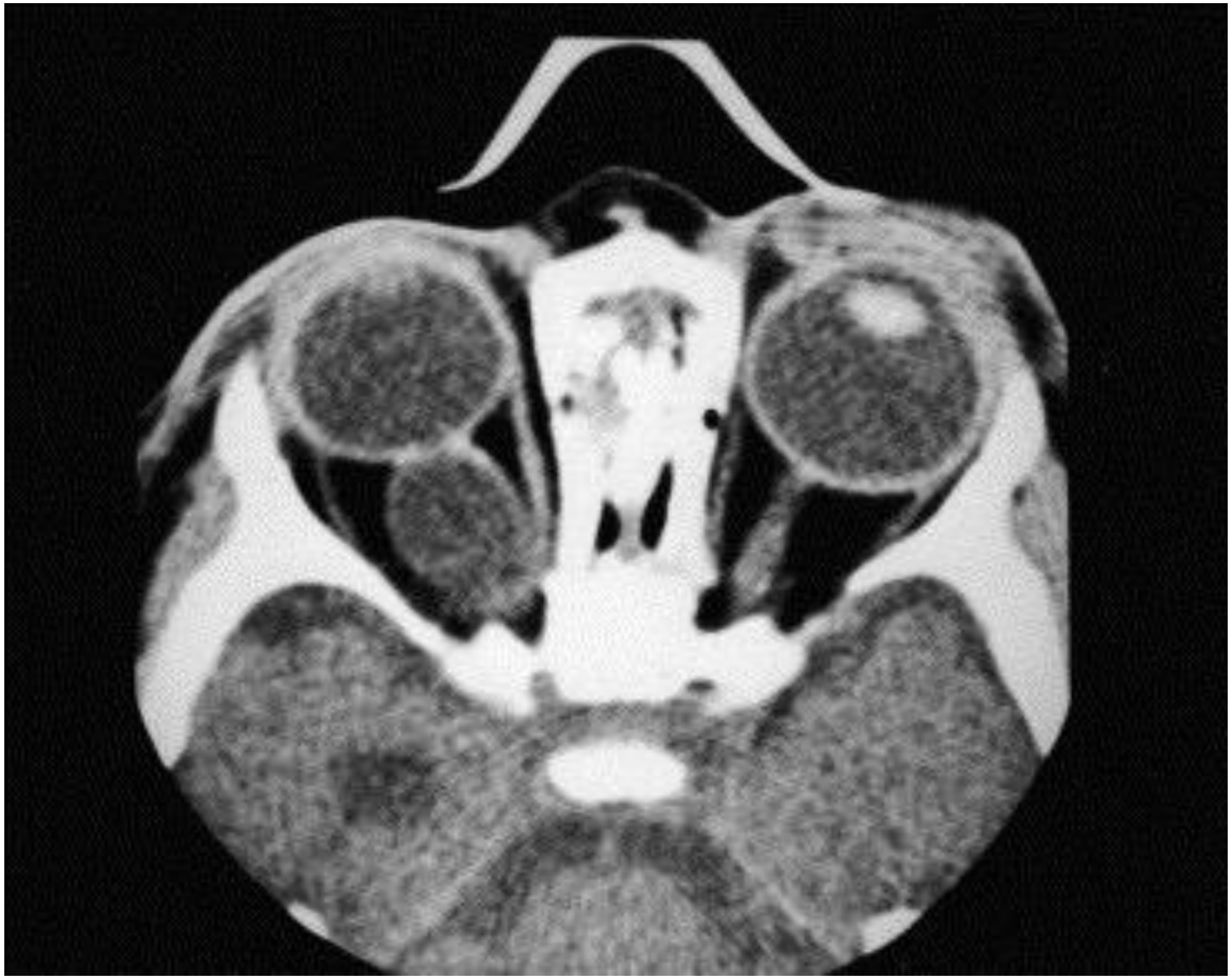
Ectropion of inner retina -
>
coloboma

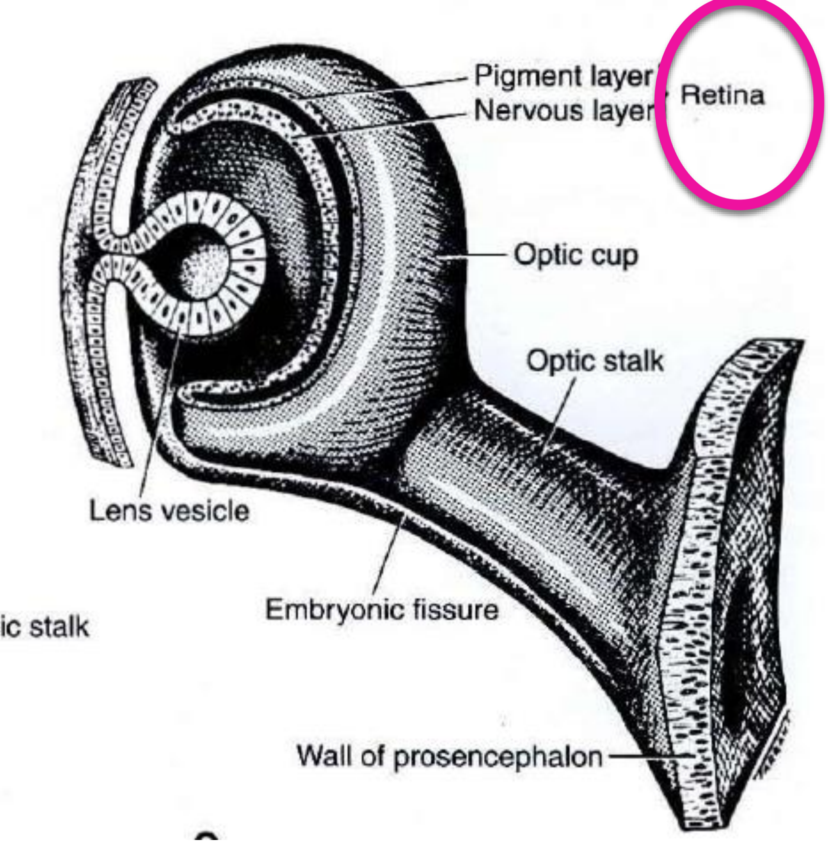


Cystic coloboma:
vesicular cavity
enlarges adjacent
to point of defect

Simple coloboma:
defective retina, RPE

Coloboma with cyst

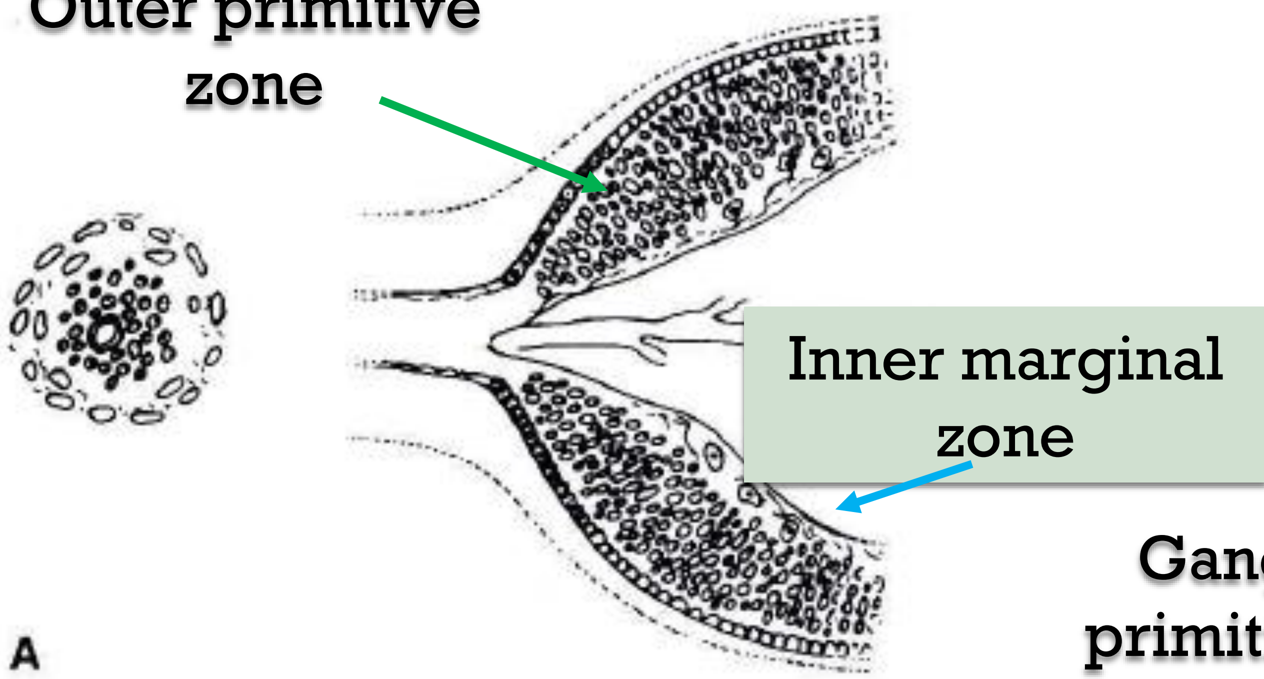




Neurosensory retina

- arises from the inner layer of neuroectodermal cells of the optic cup
- Early differentiation within 1st month, rapid mitosis -> 3-4 rows of cells
- Nuclei are densely packed in the outer 2/3 of the optic cup known as the "primitive zone"
- Inner 1/3 initially devoid of nuclei = inner marginal zone -> differentiates to the NFL

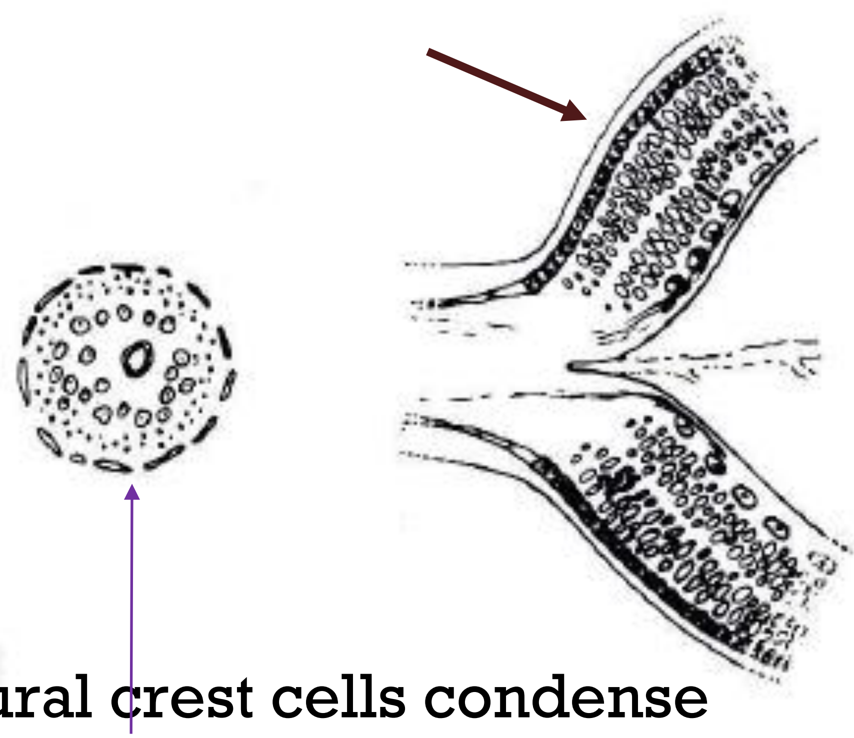
Development of the retina and optic nerve



Neurosensory retina develops from the neuroectoderm

Ganglion cells migrate from outer primitive zone to inner marginal zone

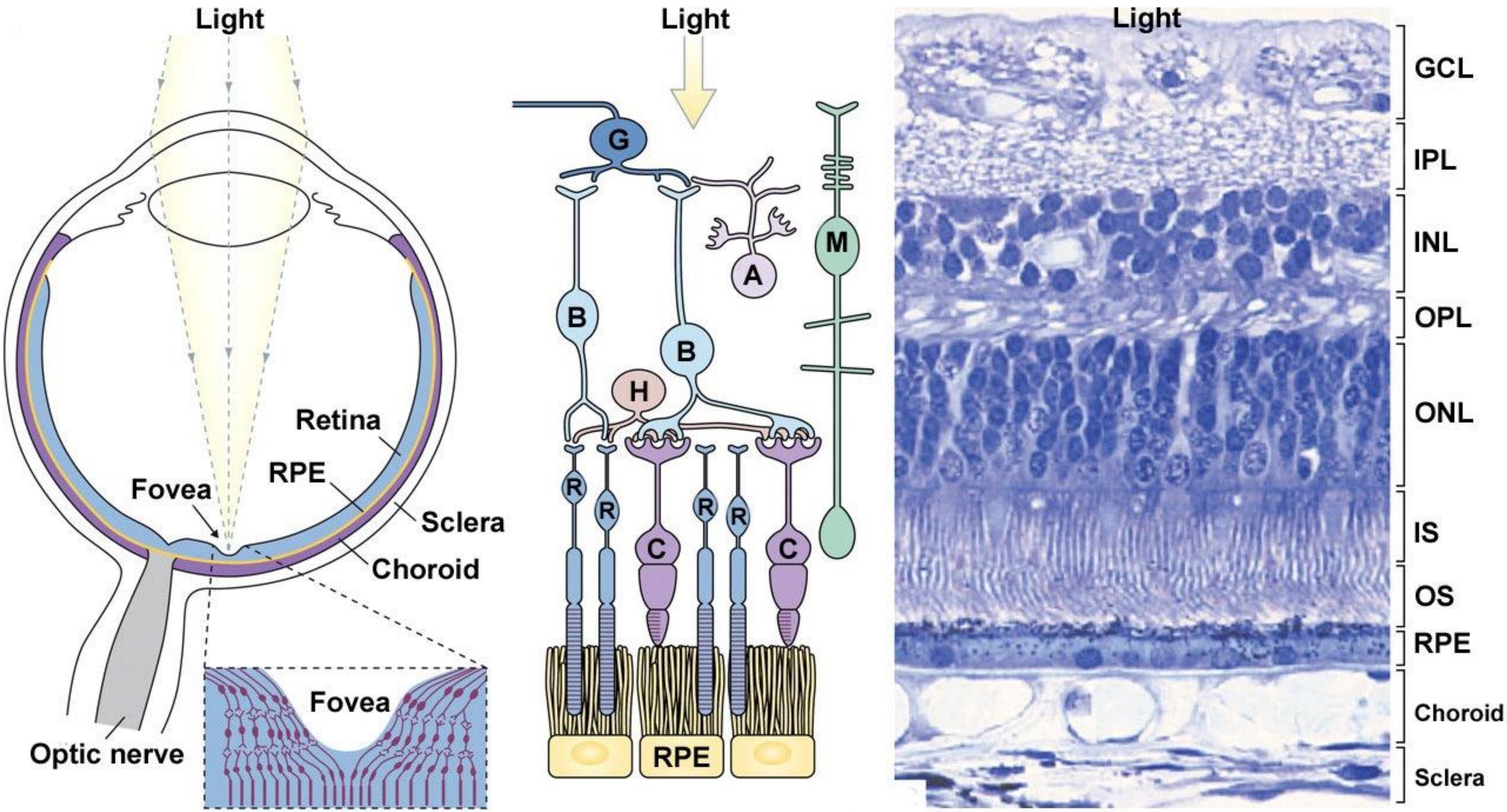
Hyaloid a. enters the vitreous (5th wk)



Migration of nuclei results in 3 nuclear and plexiform layers

Development of the retina and optic nerve

- Differentiation of the retina begins in the center of the optic cup and gradually extends peripherally toward its rim.
- ganglion and Muller cells have migrated from the outer neuroepithelial layers toward the vitreous cavity



Light
Retina
RPE
Fovea
Sclera
Choroid
Optic nerve
Fovea
Light
G
B
B
A
M
H
R
R
C
R
R
C
RPE
Light
GCL
IPL
INL
OPL
ONL
IS
OS
RPE
Choroid
Sclera

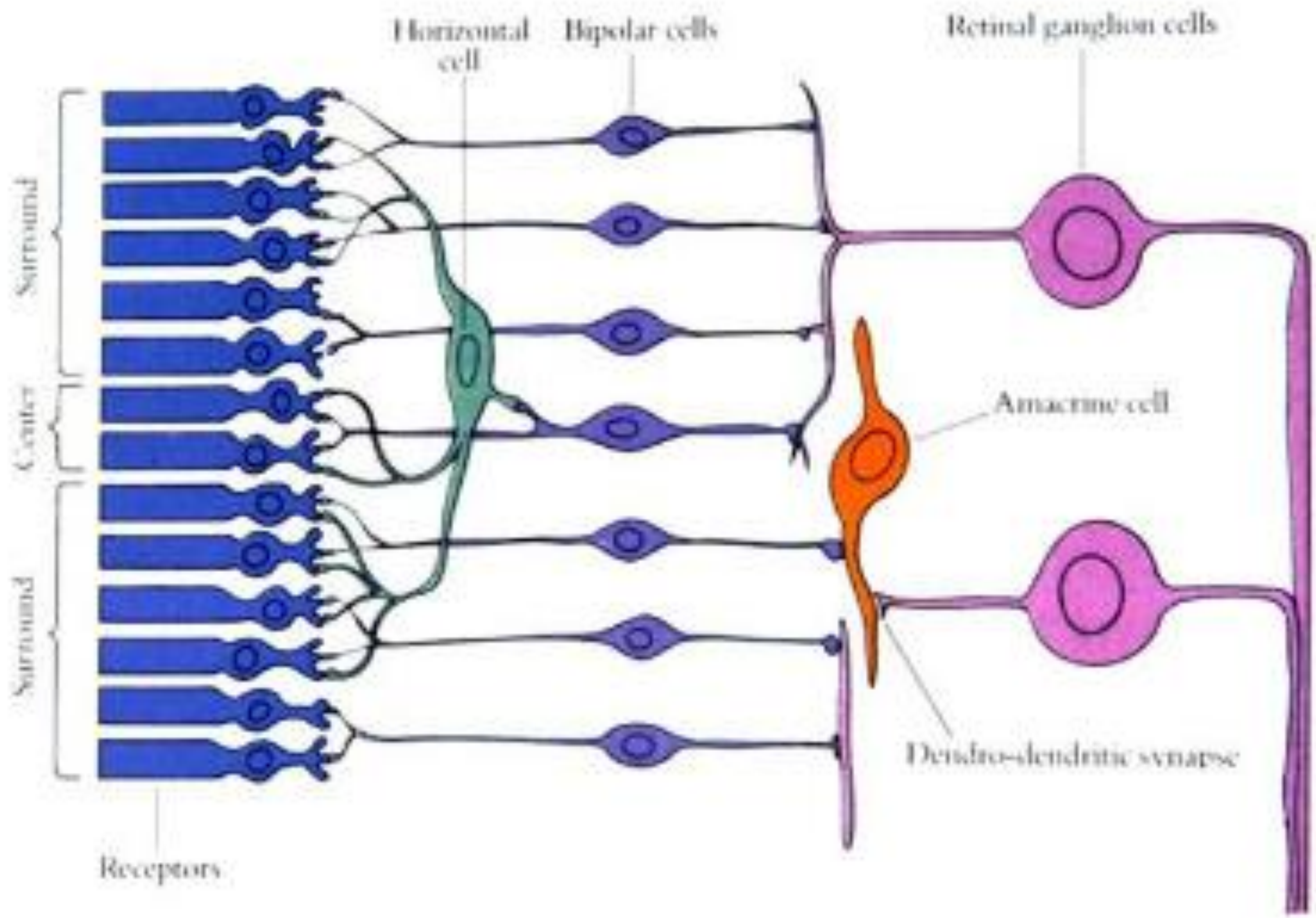
Development of the retina and optic nerve

- As a result, the nuclei of the neuroblastic cells become segregated as 2 distinct layers, the inner and outer neuroblastic layers.
- separated by a region of tangled cell processes known as the transient nerve fiber layer of Chievitz,
- which becomes the definitive inner plexiform layer between weeks 9 and 12 of gestation (except in the macula, where it persists until birth)

Development of the retina and optic nerve

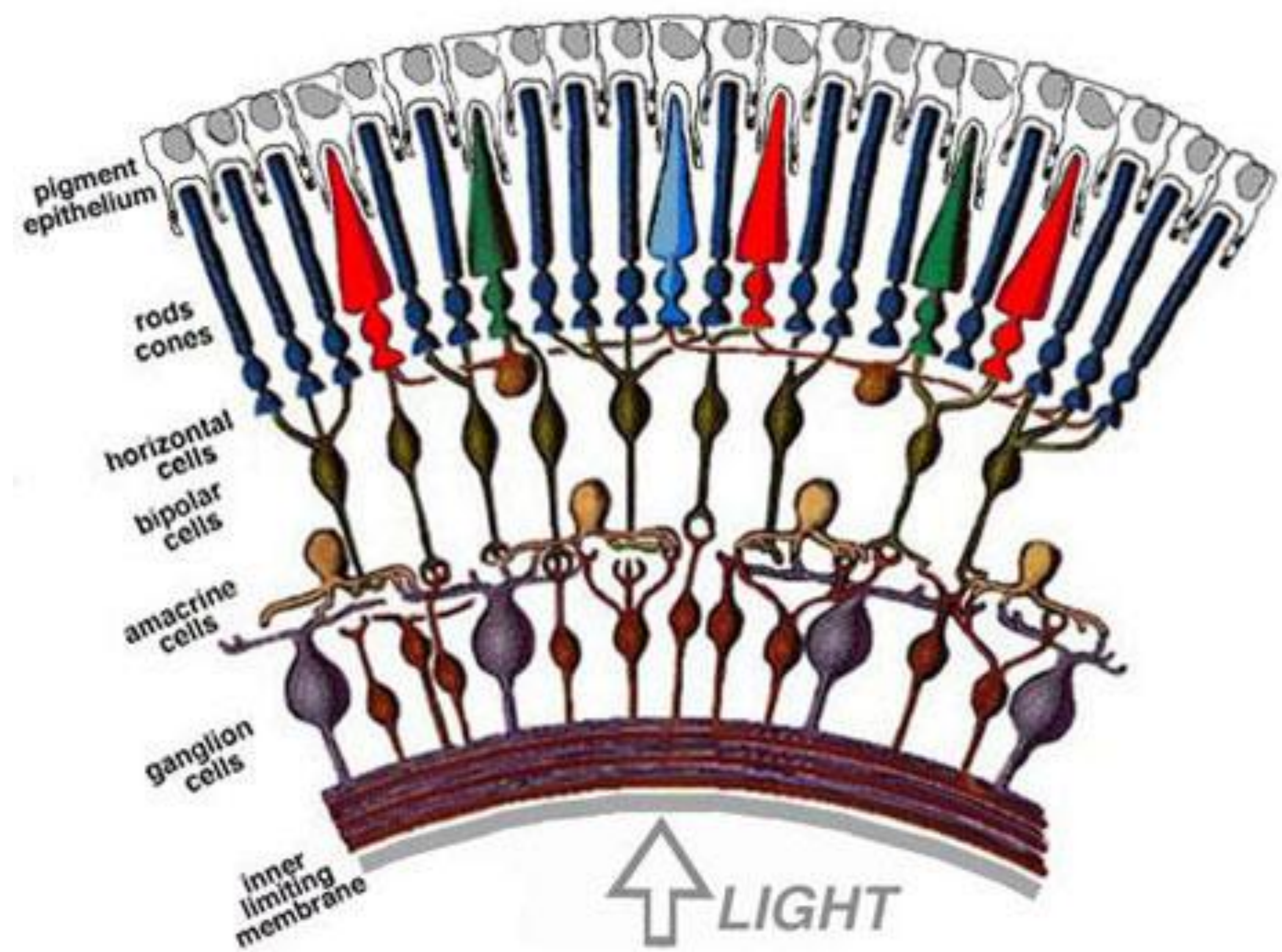
- The ganglion cells are the first cells of the retina to become clearly differentiated.
- GA 6 wk: axonal processes begin to develop
- Axons nearest the posterior pole are the first to enter the optic stalk and induce formation of the optic nerve.
- GA 15-17 wks: number of ganglion cells increases rapidly
- GA 18- 30 wk: decrease from apoptosis

Ganglion cell



Neurosensory retina

- Photoreceptors arise from the outermost layer of neuroblastic cells
- cone outer segments develop at 5 mo
- rod outer segments at 7 mo
- Amacrine cells wk. 14
- Bipolar cells wk.23 dendrites to OPL by wk 25



Fovea

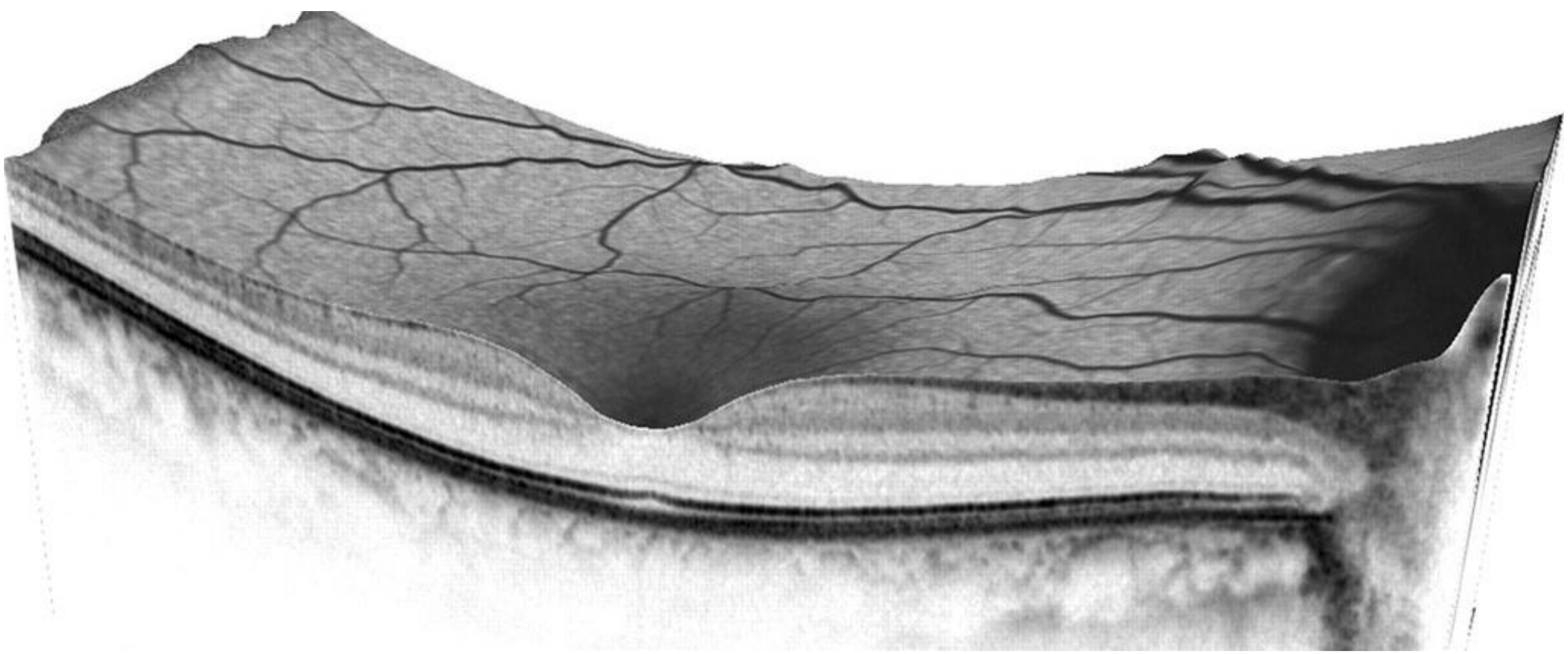
- Differentiation of the neurons, photoreceptors, and glial cells in the fovea occurs early (by wk.15)

- because this region is the focal point for the centraperipheral development of the retina

- “ จุดเริ่มต้น “

Fovea

- Thinning of the ganglion cell and inner nuclear layers
- begins at 24- 26 weeks of gestation and gives rise to the earliest recognizable depression in the area of the macula



Fovea

- Mo. 7: foveal pit more prominent:
  - marked thinning of the inner nuclear layer
  - Cone inner segments narrower and longer
  - Elongation of the Henle fiber layer

- Mo.8:
  - Only 2 layers of ganglion cells remain
  - the inner nuclear layer at the foveola is reduced to < 3 rows (lateral displacement)

Fovea

- At birth, axons of bipolar cells that pass to the inner plexiform layers constitute the prominent transient layer of Chievitz

Fovea

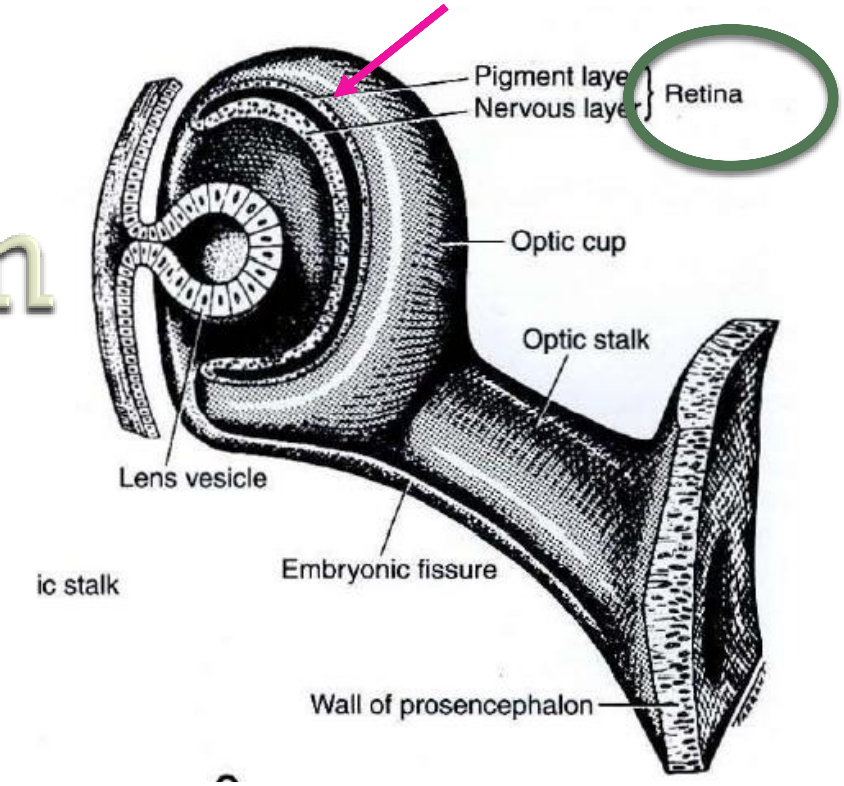
- Relocation of all layers to the periphery of the foveal slope, which leaves the nuclei of the cones uncovered in

  the foveola, occurs by 4 months after birth.

Fovea

- However, remodeling of the fovea continues until nearly 4 years of age, at which time the transient layer of Chievitz is lost completely.

Retinal pigment epithelium



- Outer layer of optic cup
- Wk 6 melanogenesis begins
- The RPE cells are the first in the body to produce melanin.
- Differentiation of the RPE begins at the posterior pole and proceeds anteriorly, so

that by 8 weeks of gestation the RPE is organized as a single layer of hexagonal columnar cells located posteriorly.

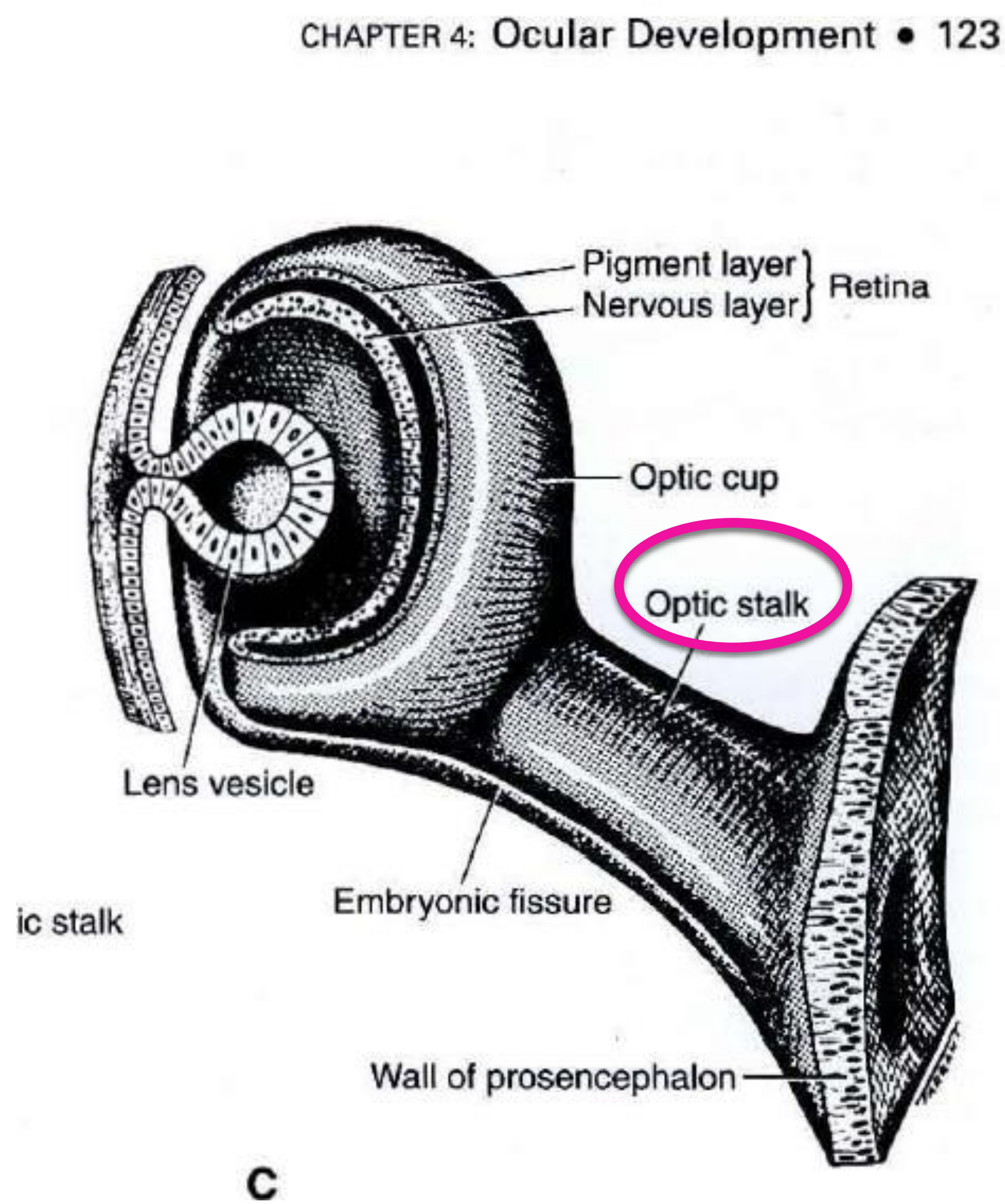
Bruch's membrane

- RPE fully functional at 3-4 mo.
- Bruch's membrane;
  - inner portion : basement membrane of RPE
  - outer layer : laid down by choriocapillaris layer

- The embryonic RPE cells have a profound inductive influence on the development of the choroid, sclera, and neurosensory retina.
- In areas where pigment epithelium does not form, the underlying choroid, sclera, and retina are hypoplastic

Optic nerve

- develops from the optic stalk, the original connection between the optic vesicle and the forebrain

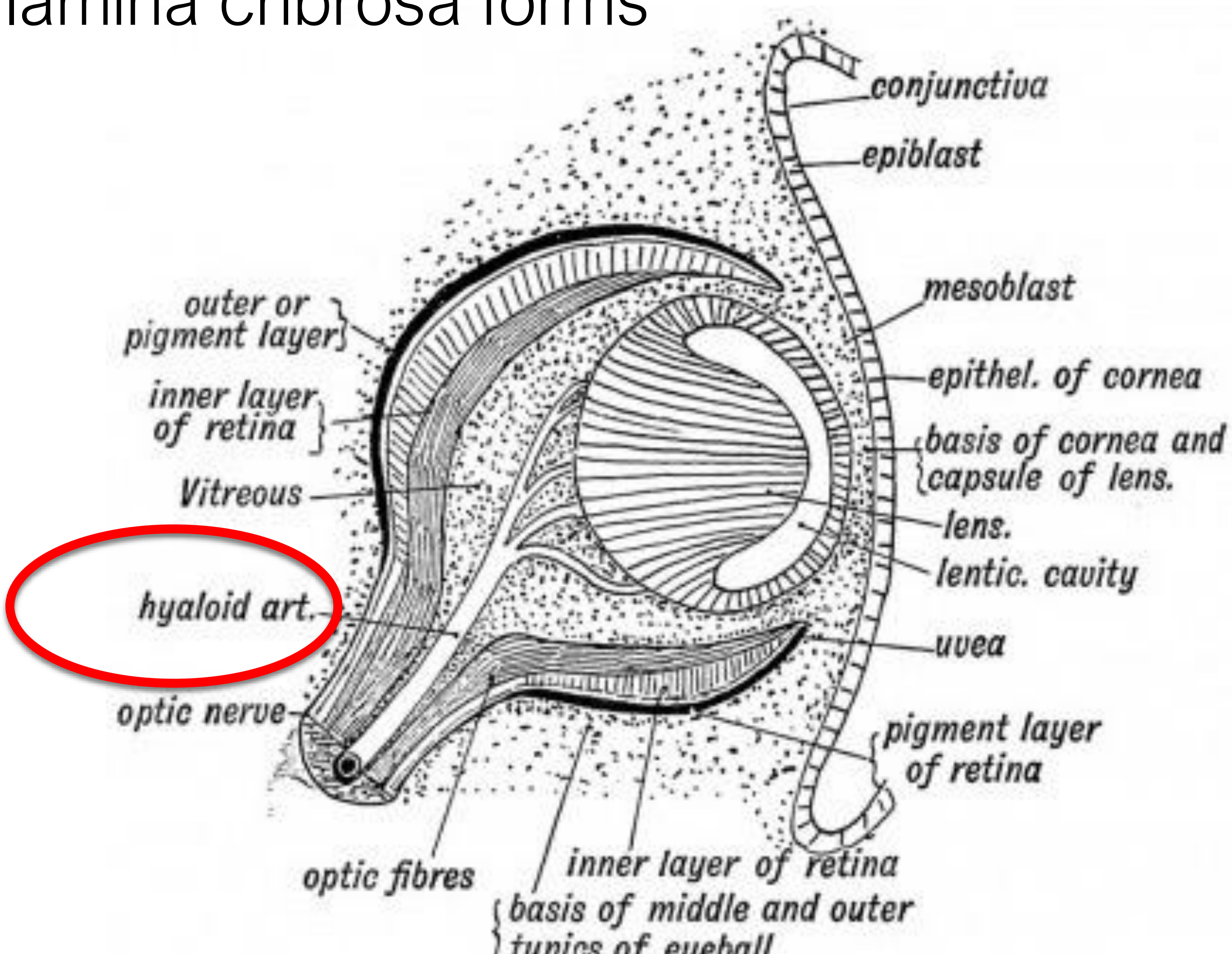


Optic nerve

- Initially, the stalk is composed
  - an inner zone of neuroectodermal cells -> degenerate, space for axons from ganglion cells to migrate into (wk.6)
  - surrounded by a layer of neural crest cells -> pia, arachnoid, and dura matter of the optic nerve (wk.7)

Optic nerve

- Wk.7: optic nerve contains hyaloid artery
- Wk.8: lamina cribrosa forms



Optic nerve

- The number of axons increases rapidly: wk.10-12 :1.9 million
- Wk. 16: rising to 3.7 million
- Later, apoptosis of axons causes the number of fibers to drop to approximately l.l million (adult) by wk 33
- Simultaneous degeneration of ganglion

  cells in the fetal retina

Optic nerve

- As the axons grow toward the lateral geniculate body, partial crossover occurs at the optic chiasm
- Myelination starts in the chiasm at mo.7, proceeds toward the eye, and ceases at the lamina cribrosa by about 1 month after birth.

Optic nerve

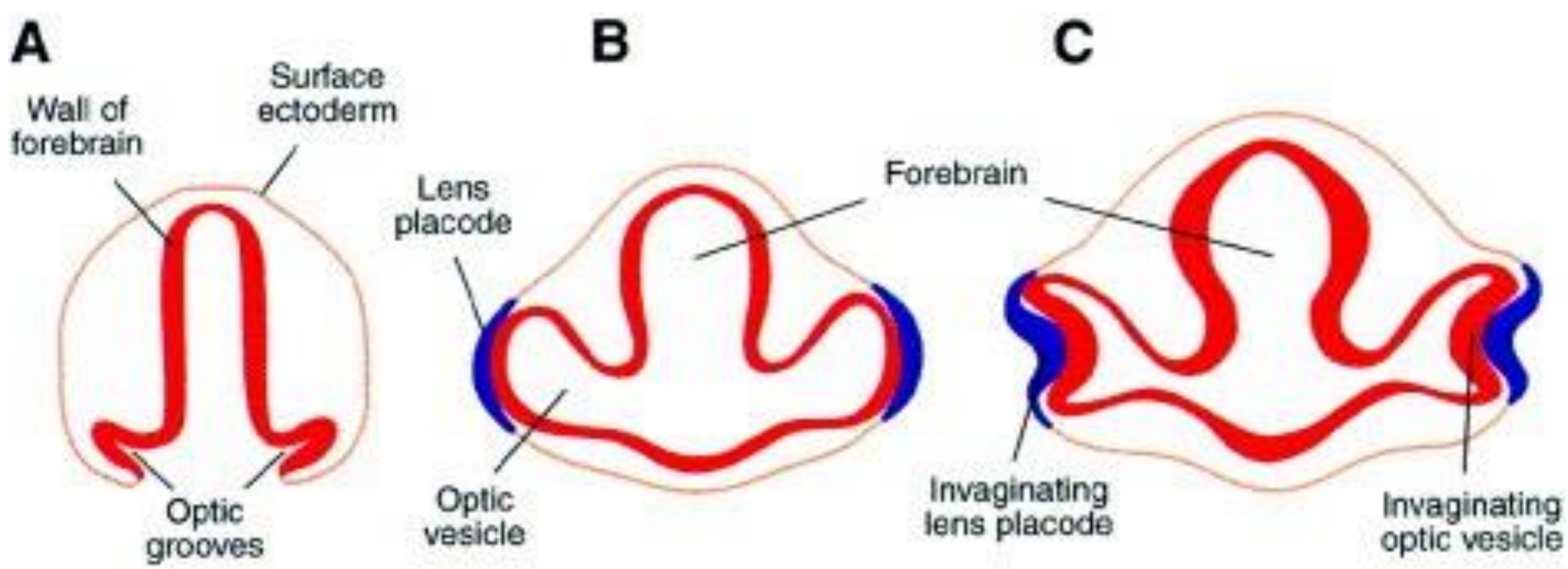
- 5 months: 50% of the growth of the optic nerve and disc has occurred
- by birth: 75%
- before 1 year of age: 95%

Lens

- The lens is first apparent at 27 days :

  disc-shaped thickening of surface epithelial cells over the optic vesicle

- lens placode invaginates to form the lens vesicle



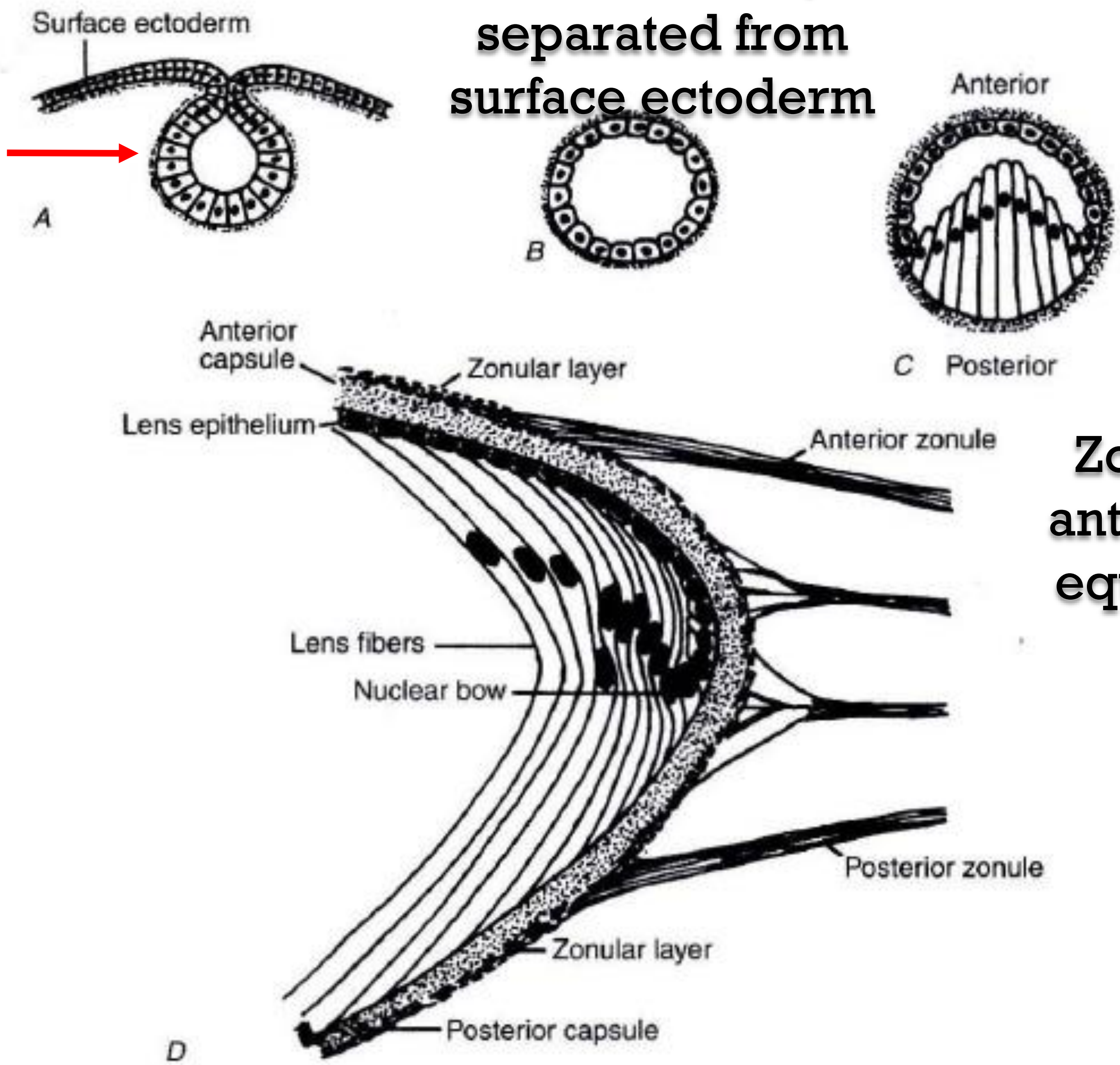
- Ultimately, the lens vesicle separates from the surface epithelium at about 33 days' gestation

Lens development

Invagination of the lens placode (surface etoderm) to form the lens vesicle

Lens vesicle, separated from surface ectoderm

Elongation of posterior cells to obliterate the vesicular cavity (mo.2 *)



Zonules attach to the anterior, posterior and equator of lens (begin wk.5)

Figure 4-10 Diagram of stages in the development of the lens and its capsule. *A*, Formation of the lens vesicle from invagination of surface ectoderm together with its basal lamina in an embryo corresponding to 32 days' gestation. *B*, Separation of the vesicle from the surface ectoderm and its surrounding basal lamina. *C*, Obliteration of the lens vesicle cavity by elongation of posterior cells at about 35 days' gestation. *D*, Equatorial region of the fully formed lens. Attachment of zonular fibers to the anterior, posterior, and equatorial regions of the lens periphery becomes apparent at approximately 5½ weeks' gestation. Note the change in polarity of cells from anterior to posterior regions of the lens. *(Modified from Tripathi BC, Tripathi BJ. Anatomy*

Lens

- Posterior cells

- Anterior (Pre-equatorial cells)

- The posterior cells of the vesicle

  account for most of the growth of the lens during the first 2 months of embryogenesis.

- The primary fibers form the compact core of the lens, known as the embryonic nucleus.

- pre-equatorial (anterior cells) cells retain their mitotic activity throughout life, producing secondary lens fibers.
- Secondary fibers form the lens sutures. The first suture marking the fetal nucleus is shaped like a Y anteriorly and an inverted Y posteriorly. (wk.7)

Lens

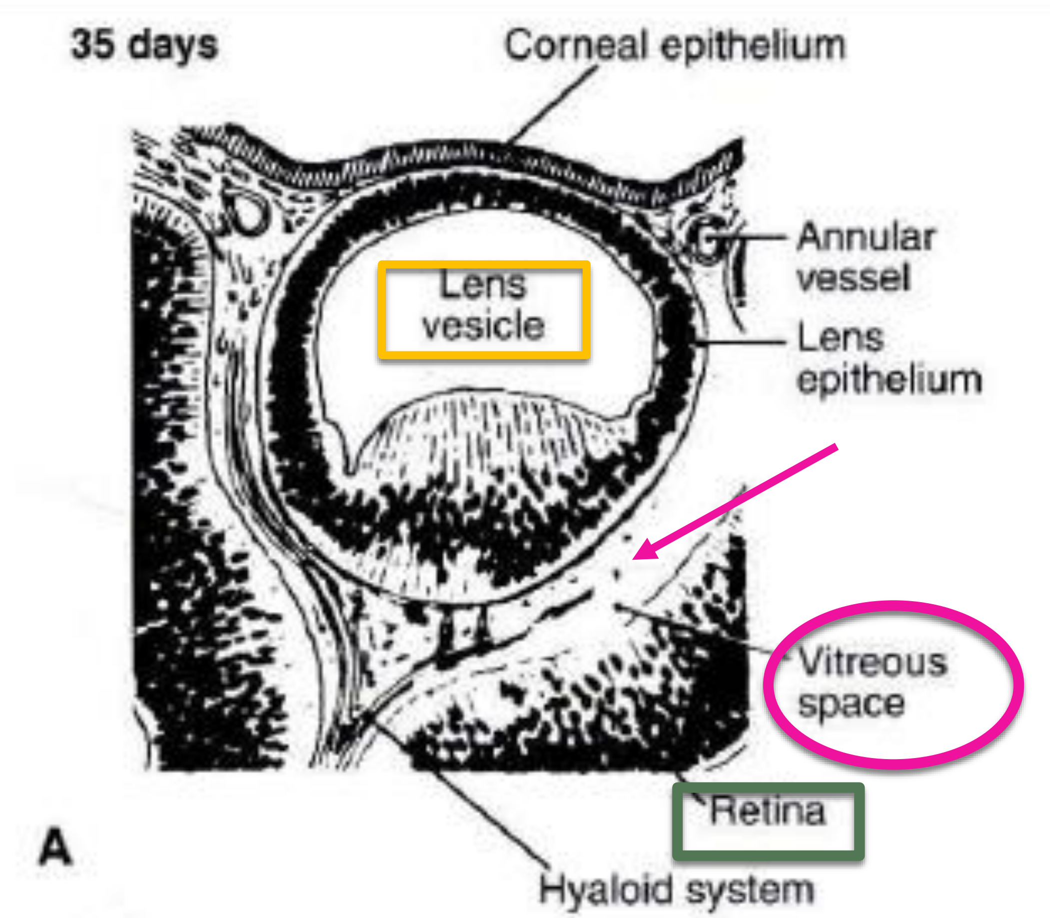
- Both the growth and the maturation of lenticular fibers continue throughout life

Vitreous

- Wk. 4-5: space between the lens vesicle and the inner layer of the optic cup becomes filled with fibrils, mesenchymal cells, and vascular channels of the hyaloid system,

the "primary vitreous"

Fully developed by mo.2



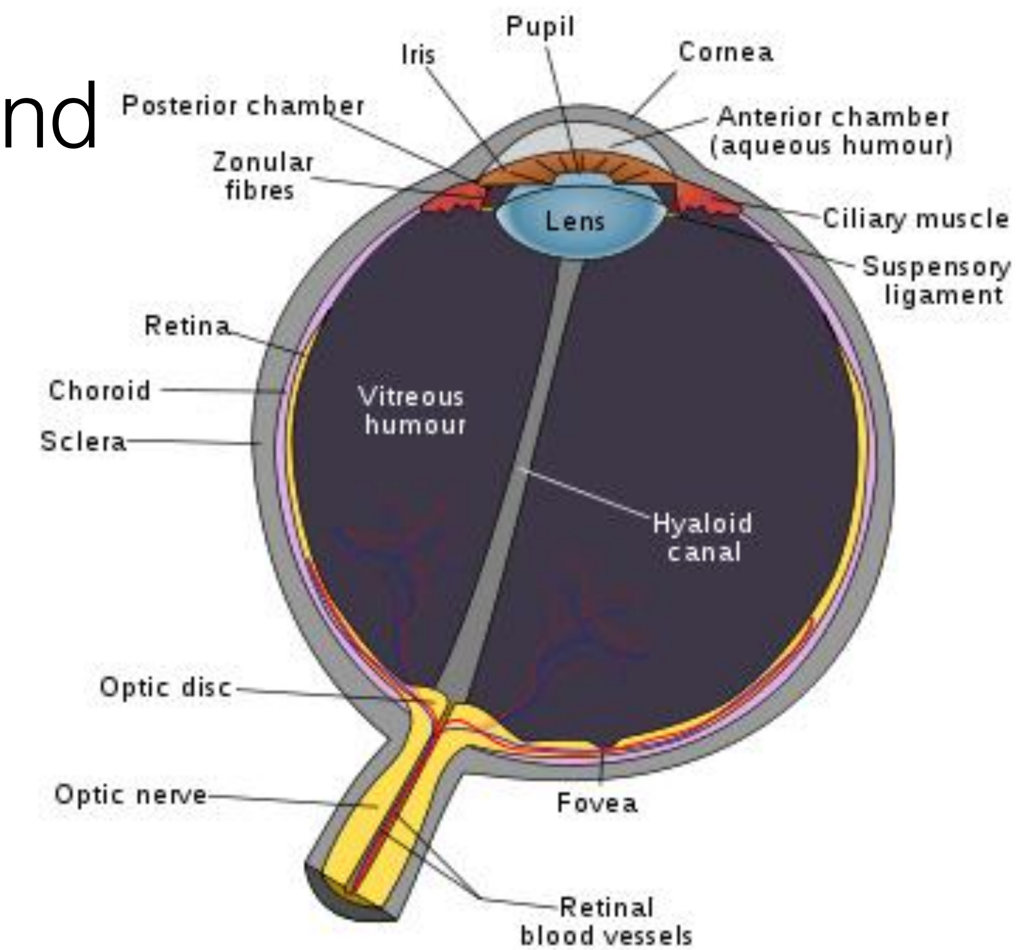
Vitreous

- development of the secondary vitreous begins soon after the primary vitreous
- secondary vitreous: avascular, type II collagen fibrils, hyalocytes

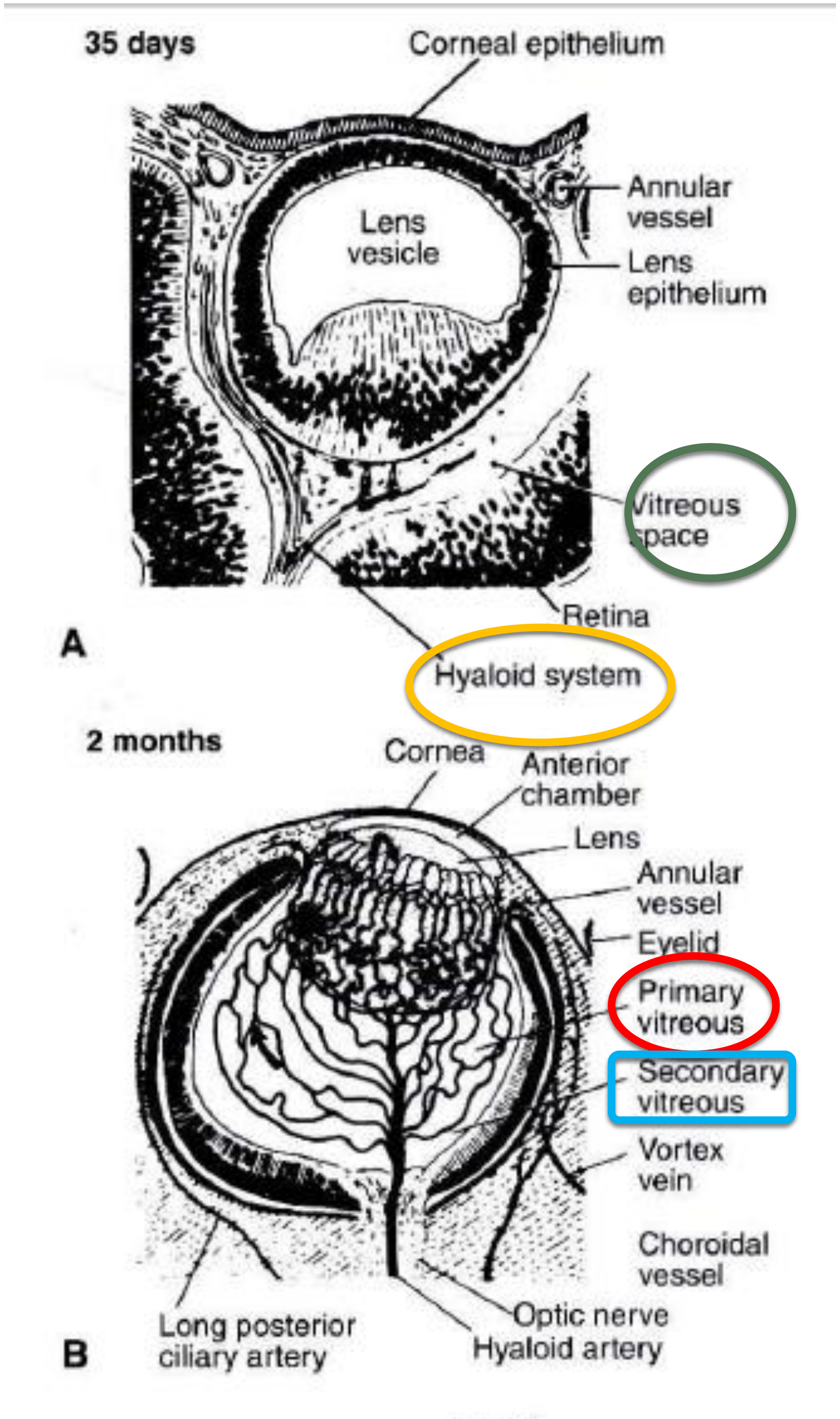
Vitreous

- Mo.3: as hyaloid system regresses and the primary vitreous retracts

- Remnants of the atrophied hyaloid system and primary vitreous remain throughout life as the Cloquet canal.



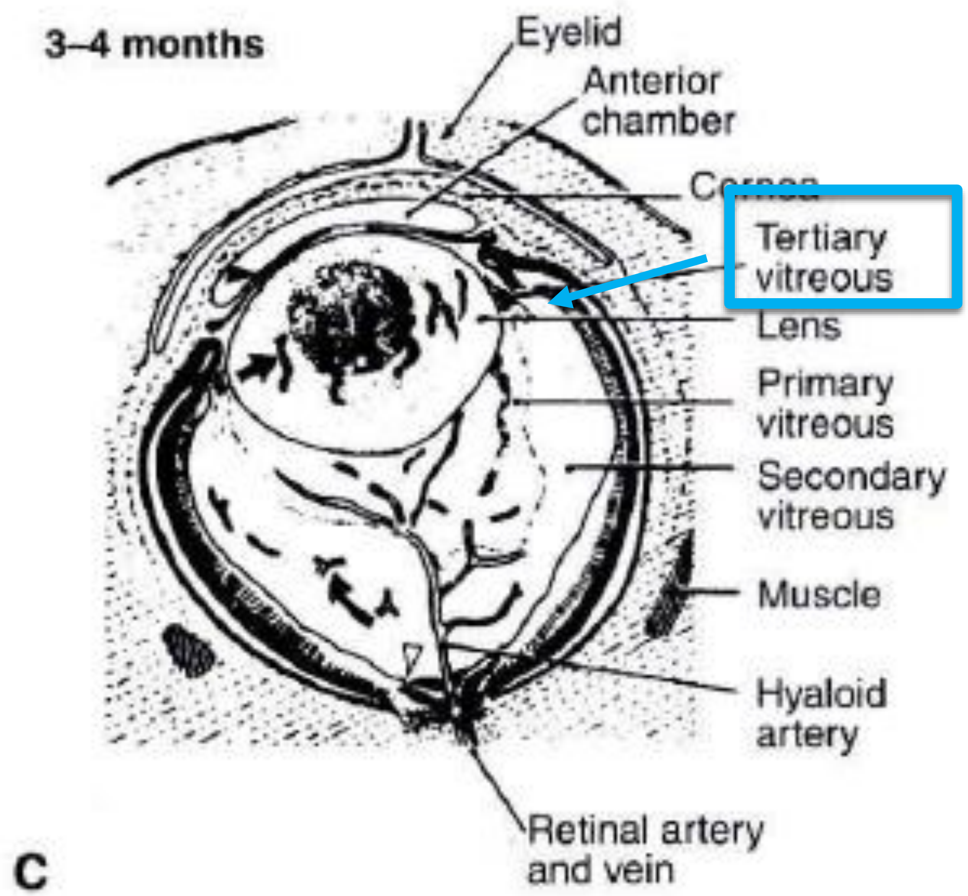
vitreous



D 35: primary vitreous and hyaloid system begin to develop behind the lens

Mo. 2: vascular primary vitreous fully developed, secondary vitreous (avascular) begins to form

vitreous



Mo. 4:
progressive atrophy of hyaloid vessels

- Tertiary vitreous (zonular fibers) stretch toward the lens capsule

Vessels through the center of the optic nerve connect with the hyaloid a.

choroid

- condensation of neural crest cells around the cup differentiate into cells of the choroidal stroma
- Wk4-5: the choriocapillaris, formed by mesodermal cells begins to differentiate

choroid

- Wk.6: primitive layer of capillaries is complete
- supplied by vessels from the internal carotid artery and the primitive ophthalmic arteries
- The channels drain into 2 main blood

  spaces, the superior orbital and inferior orbital venous plexuses -> into the cavernous sinuses

choroid

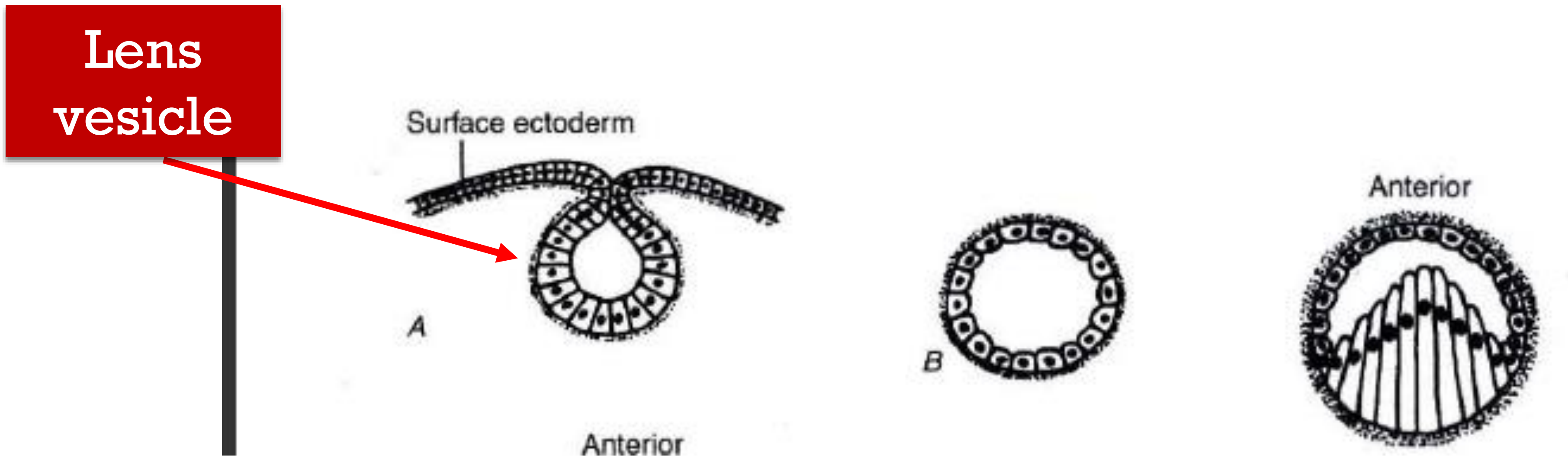
- Mo.3 : vortex viens receives efferent branches from choriocapillaris and pierce the sclera
- Mo. 5: choroidal vessels form layers

  (inner choriocapillaris , middle layer of medium-sized arterioles and the outer layer of large vessels)

choroid

- Mo.6: melanosomes appear
- Mealnocytes differentiate from neural crest cells

Cornea and sclera



- The separation of the lens vesicle from the surface ectoderm initiates the development of the cornea
- Three successive waves of ingrowth of neural crest cells associated with differentiation of the anterior chambers

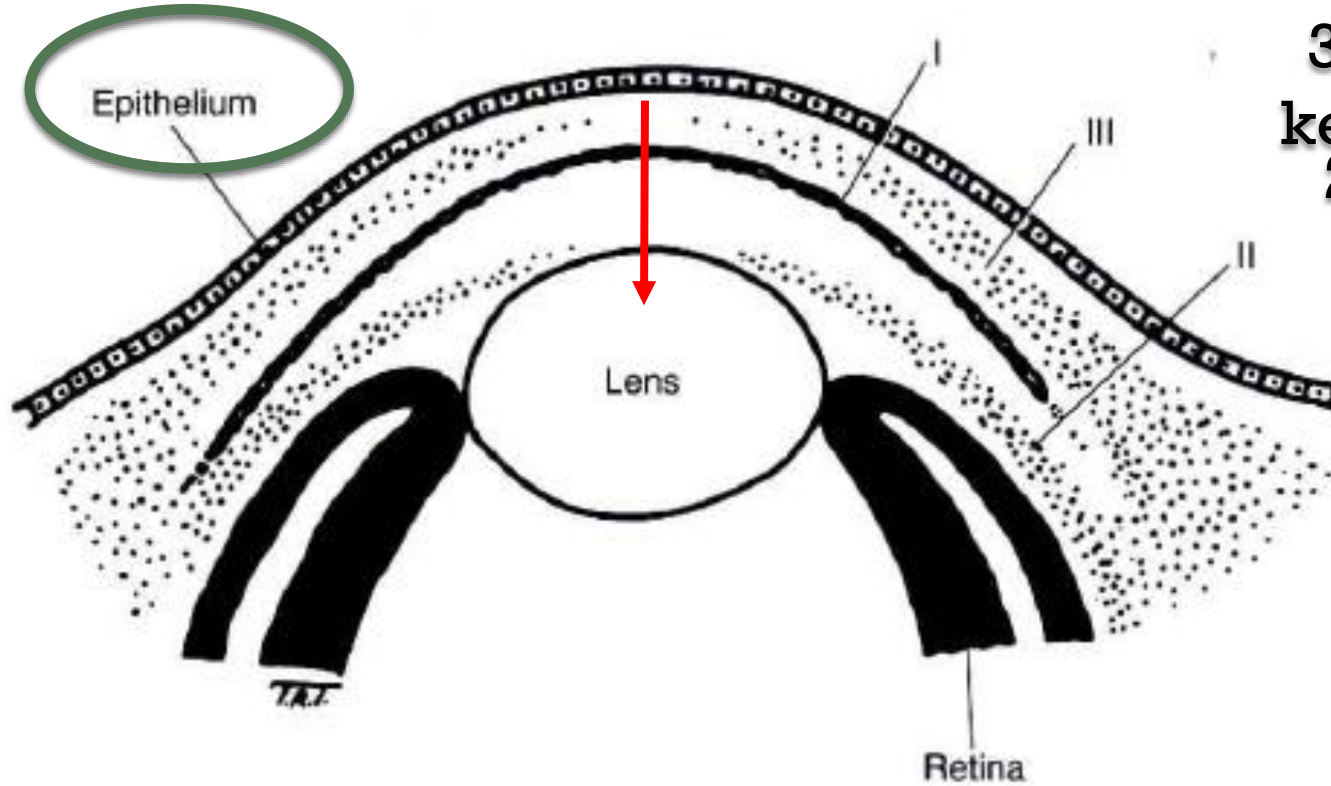
1st wave: corneal endothelium

3rd wave: keratocytes

2nd wave: iris, pupillary mb

Separation of Lens from epithelium

3 waves of neural crest cells migrate inbetween to from the corneal stroma, endothelium, iris



3 successive waves of ingrowth of neural crest cells associated with differentiation of the anterior chambers :

First wave forms the corneal endothelium

Second wave forms the iris and part of the pupillary membrane

Third wave forms keratocytes

Cornea and sclera

- At 5-6 weeks of gestation, the cornea consists of the following:
- a superficial squamous and a basal cuboidal layer of epithelial cells
- a primary stroma
- a double layer of endothelial cells posteriorly

- Wk. 7: next migratory wave of cells into the stroma ->
- differentiate into keratocytes that secrete type I collagen fibrils and form the matrix of the mature (or secondary) corneal stroma.

- The embryonic corneal stroma is approximately double the normal post-embryonic width

- Dehydration (especially of hyaluronic acid) and compression of the connective tissue cause the later reduction in thickness

- Mo. 3: endothelium becomes a single layer of flattened cells
- basal lamina of the cells -> future Descemet's membrane
- Descemet's membrane
- consists of 2 zones:
  - the lamina densa, toward the stroma
  - the lamina lucida adjacent to the endothelium

- Mo. 4: superficial keratocytes produce the Bowman’s layer

sclera

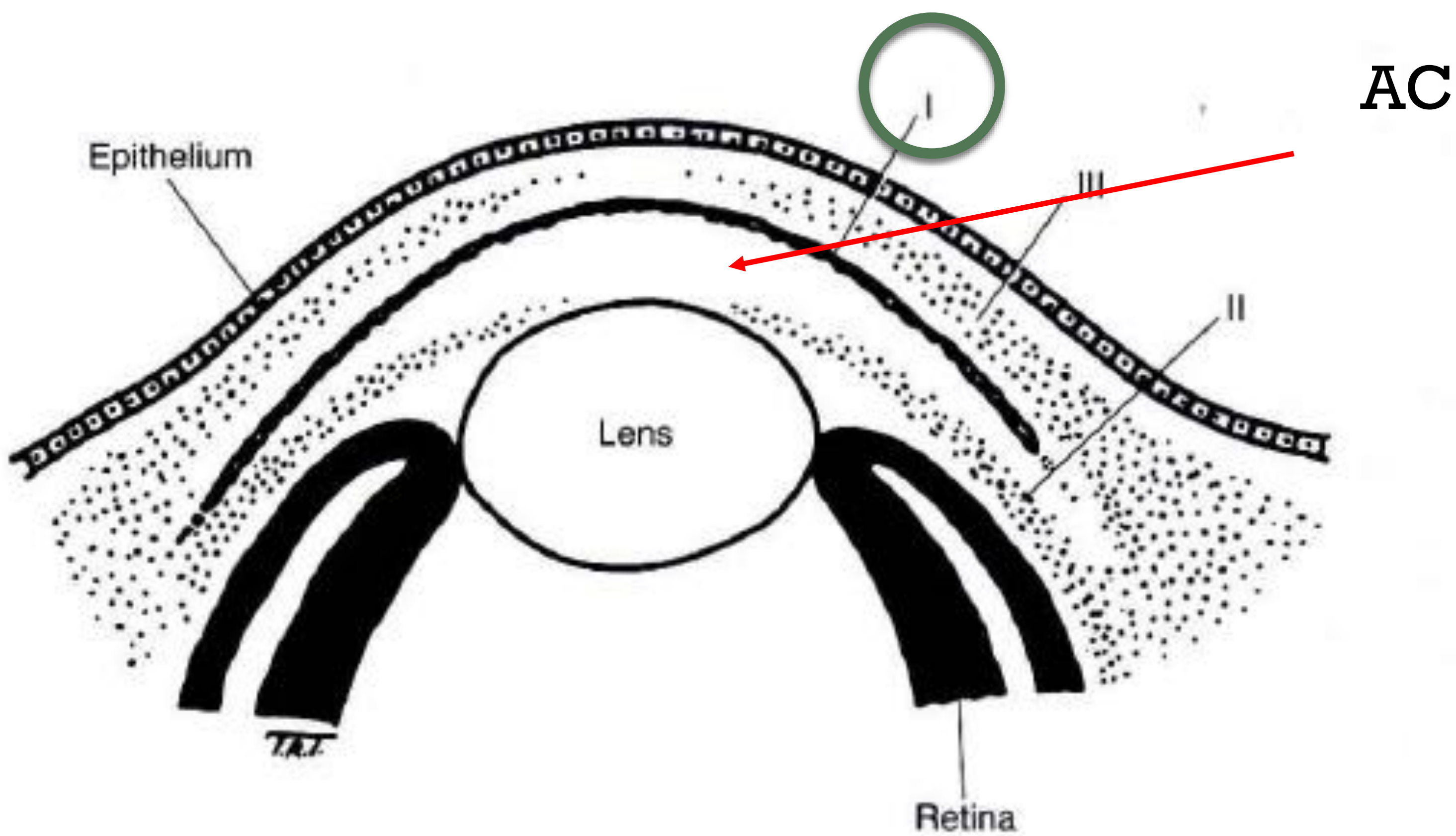
- sclera is formed by mesenchymal cells that condense around the optic cup

- Most of these cells are derived from the neural crest except the temporal sclera which is derived from the mesoderm

Anterior Chamber, Angle, Iris, and Ciliary Body

- The anterior chamber is first recognizable as the slitlike space that results after the ingrowth

  of the first wave of mesenchymal cells



Anterior Chamber, Angle, Iris, and Ciliary Body

- Wk 7: the AC angle is occupied loosely organized mesenchymal cells of neural crest origin.
- These cells will develop into the trabecular meshwork.

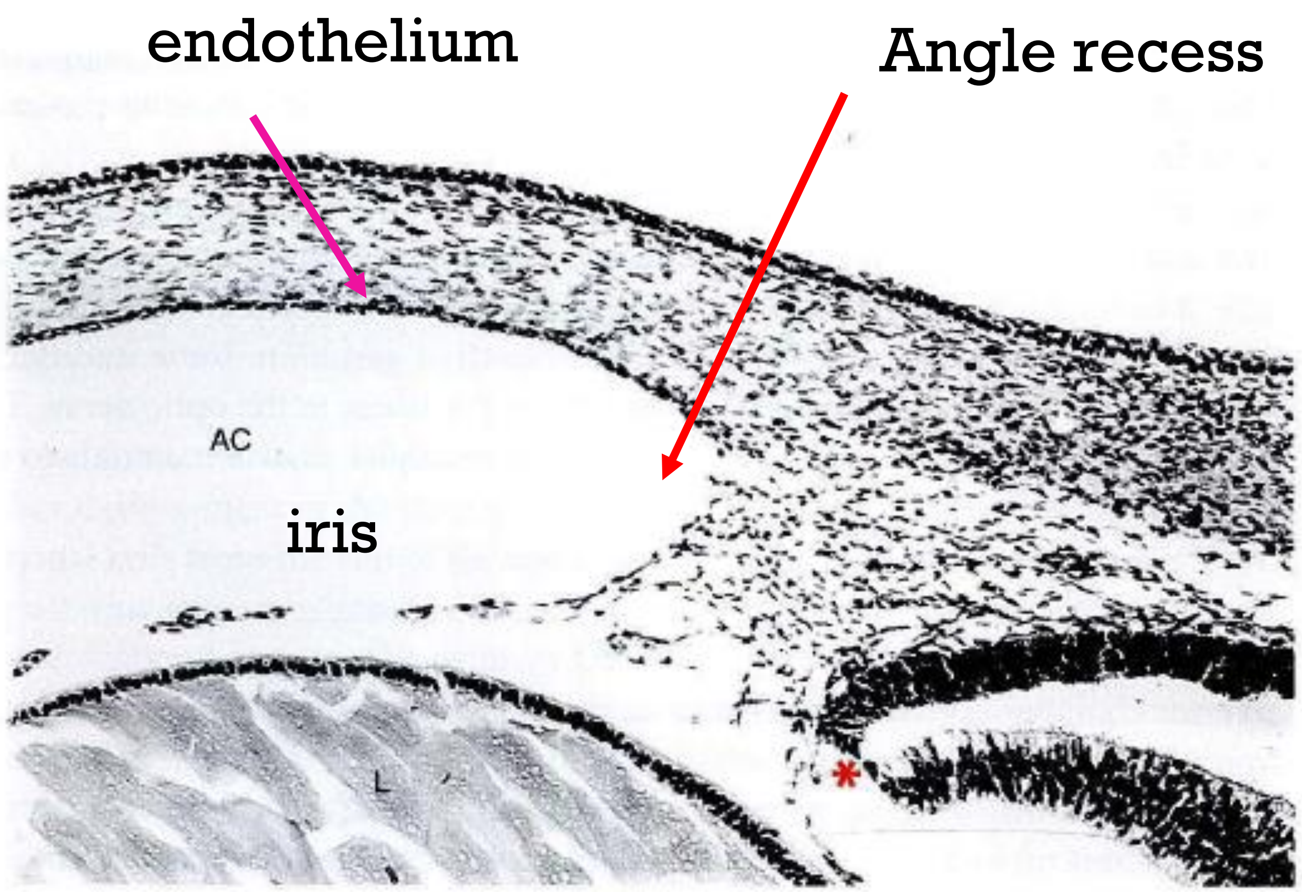


Figure 4-14 Light micrograph of the eye of an 11-week fetus in meridional section. The angular region is poorly defined at this stage and is occupied by loosely arranged, spindle-shaped cells. The Schlemm canal is unrecognizable, and ciliary muscles and ciliary processes are not yet formed; the latter are derived from neural ectodermal fold *(asterisk)*. Corneal endothelium appears continuous with the cellular covering of the primitive iris. *AC* = anterior chamber, *L* = lens. (Original magnification ×230.) *(Reproduced with permission from Tripathi RC, Tripathi BC. Functional anatomy of the anterior chamber angle. In: Tasman W, Jaeger EA, eds.* Duane's Foundations of Clinical Ophthalmology. *Philadelphia: Lippincott; 1991.)*

Wk.15:
Cells of the corneal endothelium form a layer that extends to the angle recess and meet the anterior surface of the developing iris, thus demarcating the angle of the anterior chamber

The angle recess progressively deepens until up to 4 years after birth

Anterior Chamber, Angle, Iris, and Ciliary Body

- Mo.4: Schlemm canal develops from a small plexus of venous canaliculi derived from mesoderm
- Initially function as vessels, later as aqueous sinus

Anterior Chamber, Angle, Iris, and Ciliary Body

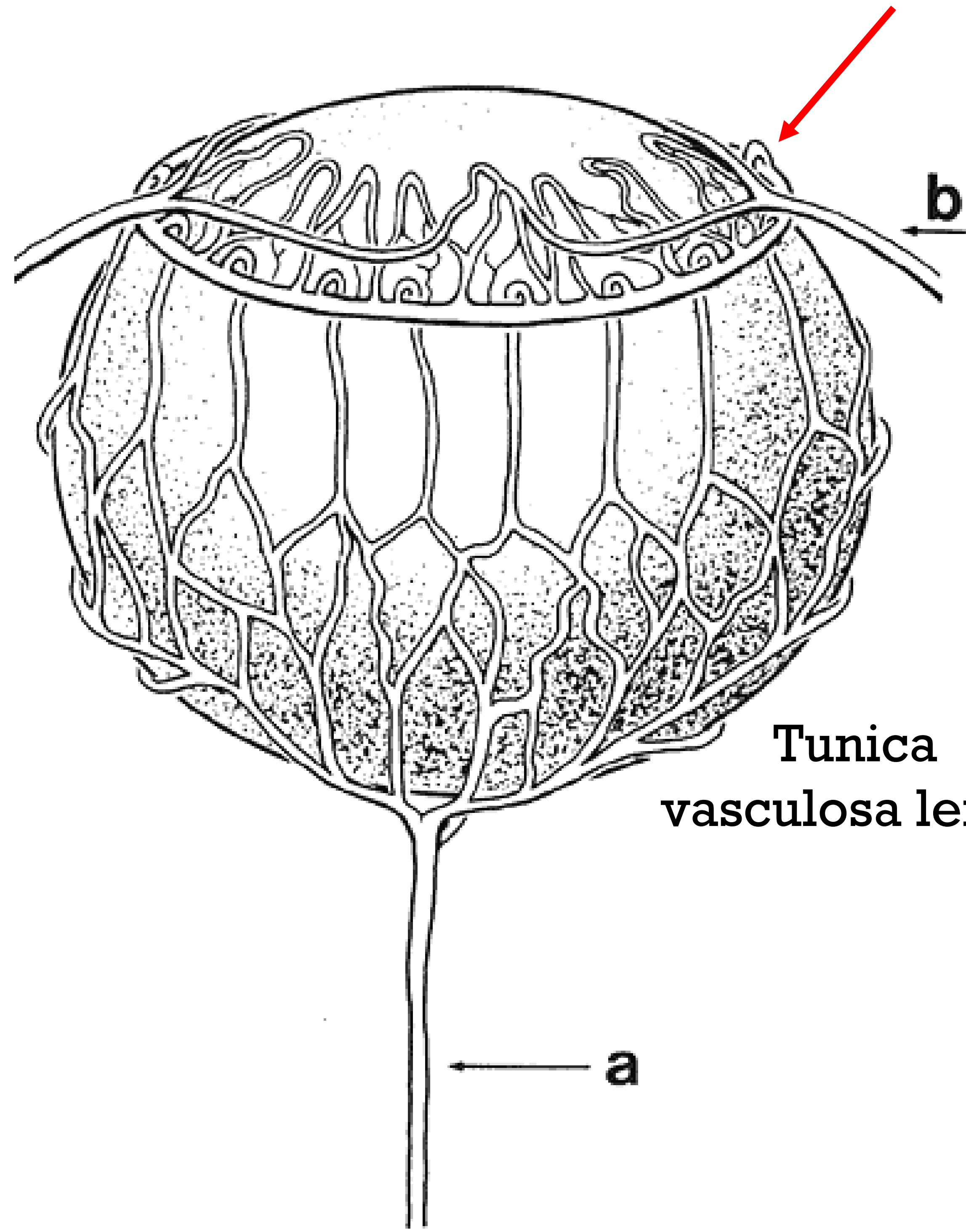
- Mo.3-4: Differentiation of the ciliary epithelium occurs in the 2 layers of neuroectoderm:
  - outer pigmented layer
  - Inner non-pigmented layer
- Mo.4: folds of the ciliary process appear

Anterior Chamber, Angle, Iris, and Ciliary Body

- ciliary muscles begin to differentiate from Mo.3
- Circular fibers continue to differentiate until 1 yr after birth

- The development of the iris is associated with the formation of the anterior portion of the tunica vasculosa lentis.

- vessels that extend onto the anterior lens surface will be incorporated into the iris stroma



Tunica vasculosa lentis

Anterior Chamber, Angle, Iris, and Ciliary Body

- The most anterior region of the tunica vasculosa lentis is replaced subsequently by the pupillary membrane

- The pupillary mb is resorbed during the 6th mo. and atrophies near term.

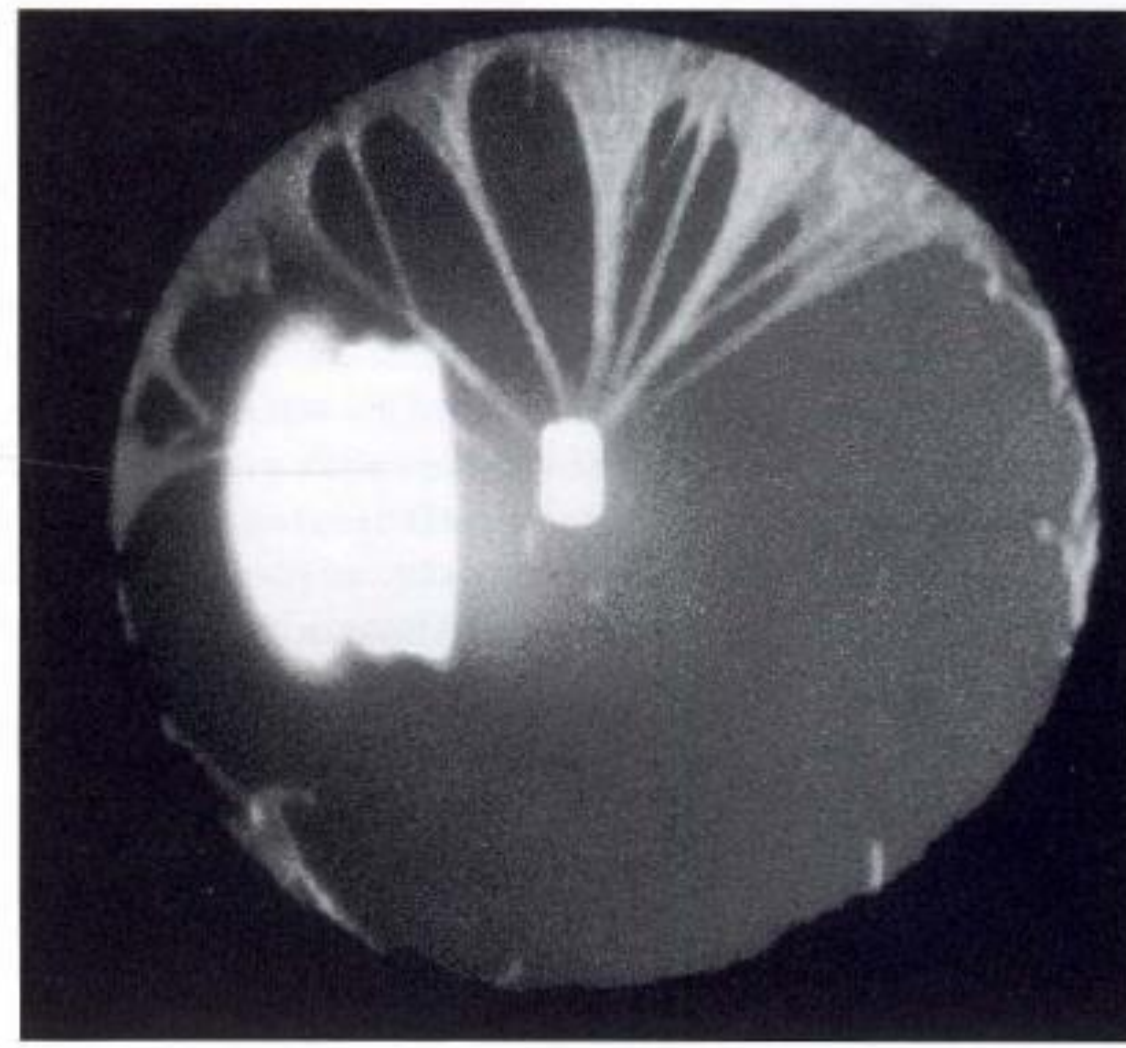
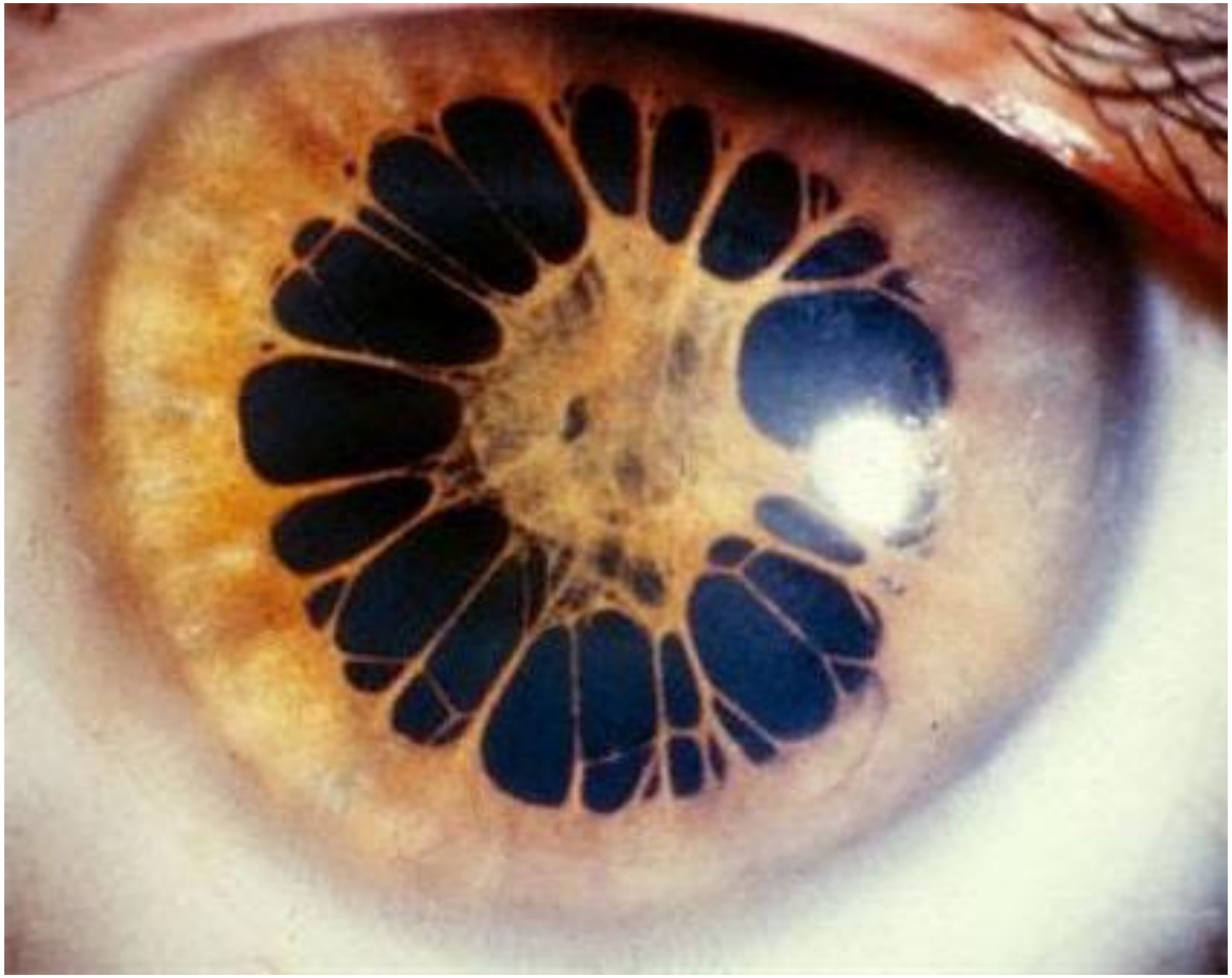


Figure 4-15 Persistent pupillary membrane.

Persistent pupillary membrane



Anterior Chamber, Angle, Iris, and Ciliary Body

- Iris sphincter muscles differentiate first (mo.3), then the dilator muscles (mo.6)
- Differentiation continues after birth
- The iris is still immature at birth.
- The collarette is closer to the pupil in the newborn eye than it is in the adult eye.

- Wk 4: The bones, cartilage, fat and connective tissues of the orbit develop from the neural crest cells that surround the optic cups

- EOM arise from mesoderm

- Wk7: the dorsomedial aspect of the superior rectus muscle gives rise to the levator muscle, which grows over the superior rectus toward the eyelid

Eyelids

- Mo.2: both the upper and the lower eyelids appear as skin folds that grow over the eye
- Mo.3: eyelids fuse
- Mo.5-7: as cilia and glandular structures develop, eyelids gradually open

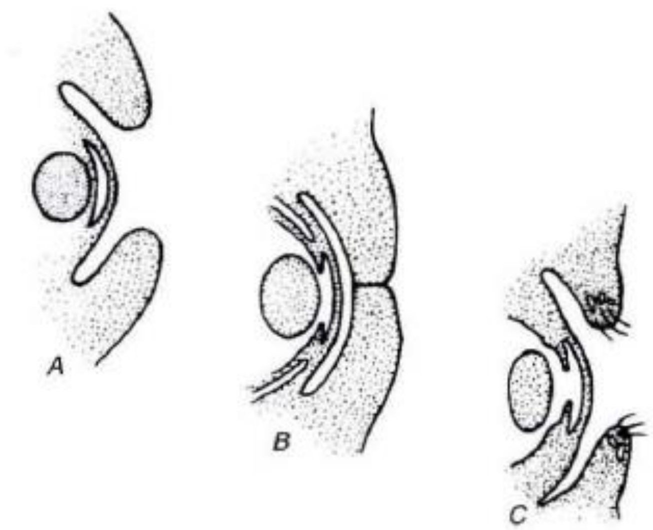


Figure 4-16 Development of the eyelids. *A*, Seventh week: upper and lower eyelid folds grow over the eye. *B*, Eyelids fuse during the eighth week; fusion starts along the nasal margin. *C*, From the fifth to the seventh month, as cilia and glandular structures develop, eyelids gradually open.

Lacrimal gland

- Wk 6-7: lacrimal glands begin to develop
- From epithelial cells (surface ectoderm)
- Mo.3: lacrimal gland ducts formed
- Lacrimal gland (reflex) tear production does not begin until 20 or more days after birth.
- Hence, newborn infants cry without tears!

Realignment of the globe

- Initially, the axes of the 2 optic cups and the optic stalks form an angle of 180

- At 3 mo.: angle has decreased to 105°

- After remodeling and repositioning:

- At birth, the axes form an angle of 71°

- However, the adult orientation of 68° is not achieved until the age of 3 years.

Development also influenced by:

- genetic factors
- Non-genetic teratogens:
  - toxins
  - maternal infections
  - nutritional deficiencies
  - Radiation
  - Drugs
  - developmental failures
  - traumatic insult

Thank you for your attention